Representation Theory (Appendix)

Appendix A

Representation Theory

A.1 Abstract Representation Theory

A.1.1 Semisimple Rings

Definition A.1.1. *Let R be a ring. A left/right R-module is a commutative group M with a biadditive map* $\begin{array}{ccc} R \times M & \to & M \\ (a,x) & \mapsto & ax/xa \end{array}$ *s.t.*

$$\begin{cases} 1x = x, \ \forall x \in M & x1 = x, \ \ \forall x \in M \\ (ab)x = a(bx), \ \forall a,b \in R, \forall x \in M & x(ab) = (xa)b, \ \forall a,b \in R, \forall x \in M \end{cases}.$$

$\to$ if the ring R is commutative, the notions of left and right R-module coincide.

e.g. the ring R with left/right multiplication by itself is a left/right R-module, called the **left/right R-module.**

Definition A.1.2. *Let M be a R-module. A R-submodule of M is a subgroup N of M s.t. $ax \in N, \ \forall a \in R, \ x \in N$.*

e.g. a submodule of the left R-module $M = R$ is just a left ideal of R.

Definition A.1.3. *If M is a R-module and N is a submodule of M, then the quotient group M/N has a structure of R-module given by $a(x+N) = ax+N, \ \forall a \in R, \ x \in M$. This is called a quotient R-module.*

Definition A.1.4. *Let M be a R-module and $(M_i)_{i\in I}$ be a family of submodules of M. Then M is the sum of the M_i and write $M = \sum_{i\in I} M_i$ if $\forall x \in M, \ \exists x_i \in M_i$, all but finitely many equal to 0, s.t. $x = \sum_{i\in I} x_i$. Additionally M is the direct sum of M_i and write $M = \bigoplus_{i\in I} M_i$ if $\forall x \in M, \ \exists! x_i \in M_i$ s.t. $x = \sum_{i\in I} x_i$.*

Definition A.1.5. *Let M, N be R-modules. A R-linear map from M to N is a morphism of abelian groups $\phi : M \to N$ s.t. $\phi(ax) = a\phi(x), \ \forall a \in R, x \in M$.*

Definition A.1.6. *An exact sequence of R-modules is an exact sequence of abelian groups where all the abelian groups are R-modules and all the maps are R-linear.*

e.g. for $\phi : M \to N$ a R-linear map, $0 \to \ker\phi \to M \to N \to \mathrm{coker}\phi \to 0$ is an exact sequence of R-modules.

Definition A.1.7. *Let R be a ring and M be a R-module. Then M is simple or irreducible if $M \neq 0$ and if the only R-submodules of M are 0 and M. Additionally M is said to be semi-simple or completely reducible if for every submodule N of M, there exists a submodule N' of M s.t. $M = N \oplus N'$.*

Lemma A.1.1. *If $M \neq 0$ and M is semisimple, then it has a simple submodule.*

pf. Let $x \in M \setminus \{0\}$ and set $M' = Rx$. Then M' is semisimple and nonzero, so we may assume $M = M'$. Let X be the set of submodules N of M s.t. $x \notin N$, ordered by inclusion. This set is nonempty because it contains the zero submodule. If $Y \subset X$ is totally ordered, $N = \cup_{N' \in Y} N'$ is a submodule of M. Also $x \notin N$ by definition, so N is an element of X, so $\exists$ an upper bound of any totally ordered subset of X. Then by Zorn's lemma, X has a maximal element N. As $x \notin N$, $N \neq M$. As M is semisimple, there exists a submodule N' of M s.t. $M = N \oplus N'$ and $N' \neq 0$ because $N \neq M$. If N' is not simple, choose a submodule $0 \neq N'' \subsetneq N'$, so that $N \oplus N'' \notin X$ by maximality of N in X, so $x \in N \oplus N''$, but then $N \oplus N'' = M$ a contradiction. Thus N' is simple and the lemma follows.

Theorem A.1.1. *Let R be a ring and M be a R-module. Then TFAE*

1. *M is semisimple*
2. *M is the direct sum of a family of simple submodules*
3. *M is the sum of a family of simple submodules*

pf.

- $2 \to 3$ direct sums are sums
- $1 \to 3$

  assume M is semisimple. Let M' be the sum of all the simple submodules of M and choose a submodule M'' of M st. $M = M' \oplus M''$. If $M' \neq M$, $M'' \neq 0$, so by the previous lemma, M''has a simple submodule N, but then $N \subset M'$, a contradiction that M', M'' are in direct sum. So $M' = M$.

- $3 \to 1$

  let (M_i) be the family of all simple submodules of M. Assume $M = \sum_{i \in I} M_i$. Let N be a submodule of M. Define the set X be a subset K of I s.t. the number $N + \sum_{k \in K} M_k$ is direct, ordered by inclusion. This set is not empty since $\emptyset \in X$. If $Y \subset X$ is a totally ordered subset, let $K = \cup_{K' \in Y} K'$. Let $n \in N$ and $(m_k)_{k \in K} \in \Pi_{k \in K} M_k$ s.t. $n + \sum m_k = 0$. Let $K_0 \subset K$ be a finite subset s.t. $m_k = 0, \ \forall k \in K \setminus K_0$. Then $\forall k \in K_0, \ \exists L_k \in Y$ s.t. $k \in L_k$. As Y is totally ordered and K_0 is finite, $\exists L \in Y$ s.t. $\cup_{k \in K_0} L_k \subset L$. Then $L \in X$, so $n + \sum m_k = 0$ implies $n = 0$ and $m_k = 0, \ \forall k \in K_0$. By the choice of K_0, we get n and all m_k are 0. So $K \in X$. Then by Zorn's lemma X has a maximal element J. Let $M' = N \oplus \bigoplus_{j \in J} M_j$. Let $i \in I$. If $M_i \not\subset M', \ M' \cap M_i = 0$ given M_i being simple. Then the sum $M' + M_i = (N + \sum_{j \in J} M_j) + M_i$ is direct, so $J \cup \{i\} \in X$, a contradiction to the maximality of J, so $M = M'$, so M is semisimple.

Definition A.1.8. *Let R be a ring and M be a R-module. Then M is said to be finitely generated if there exists a finite family $(x_i)_{i \in I}$ of M s.t. $M = \sum_{i \in I} Rx_i$. We say that M is cyclic, if $\exists x \in M$ s.t. $M = Rx$.*

Theorem A.1.2. *Let R be a ring. TFAE*

1. *all s.e.s. of R-modules split*
2. *all R-modules are semisimple*
3. *all finitely generated R-modules are semisimple*
4. *all cyclic R-modules are semisimple*
5. *the left regular R-module R is semisimple*

pf.

- $1 \leftrightarrow 2$ by the definition of semisimple
- $2 \to 3 \to 4 \to 5$ is obvious
- $5 \to 2$

  let M be a R-module. If $x \in M, \ Rx \subset M$ is a quotient of R, so it is semisimple because R is semisimple. As $M = \sum_{x \in M} Rx$, by the previous theorem, M is semisimple.

Definition A.1.9. *A R-module P is called projective if for every surjective R-linear map $\pi : M \to N$ and every R-linear map $\phi : P \to N$, there exists a R-linear map $\psi : P \to M$ s.t. $\pi\psi = \phi$.*

$$\begin{tikzcd} & P \arrow[d, "\phi"] \arrow[dl, dotted, "\psi"'] \\ M \arrow[r, "\pi"] & N \end{tikzcd}$$

Definition A.1.10. *A R-module F is called free if there exists a family $(e_i)_{i\in I}$ of elements of F s.t. $F = \bigoplus_{i\in I} Re_i$ and s.t. $\forall i \in I$, the map $\begin{array}{ccc} R & \to & Re_i \\ a & \mapsto & ae_i \end{array}$ is an isomorphism.*

Lemma A.1.2. *Let P be a R-module. Then TFAE*

1. *P is projective*
2. *P is a direct summand of a free R-module*

pf.

- $2 \to 1$

  every free R-module is projective.
  pf. if $F = \bigoplus Re_i$, let $\pi : M \to N$ be a surjective R-linear map and $\phi : P \to N$ be a R-linear map. $\forall i \in I$ choose $x_i \in M$ s.t. $\pi(x_i) = \phi(e_i)$. Define a R-linear map $\begin{array}{ccc} \psi : P & \to & M \\ e_i & \mapsto & x_i \end{array}$. Then $\pi\psi = \phi$ and it is projective.
  a direct summand of projective module is projective
  pf. let P be projective and suppose $P = P' \oplus P''$. Let $\pi : M \to N$ be surjective R-linear map and $\phi' : P' \to N$ be a R-linear map. Let $\phi = \phi' + 0 : P = P' \oplus P'' \to N$. As P is projective, $\exists \psi : P \to M$ s.t. $\pi\psi = \phi$. Write $\psi = \psi' \oplus \psi''$ where $\begin{cases} \psi' : P' \to M \\ \psi'' : P'' \to M \end{cases}$. Then $\pi\psi' = \phi'$ and the stmt follows.
  Then free module is a projective module and its direct summand is also projective and the lemma follows.

- $1 \to 2$

  suppose P is projective. Let $M = \bigoplus_{x\in P} R$ and define $\begin{array}{ccc} \phi : M & \to & P \\ (a_x)_{x\in P} & \mapsto & \sum_{x\in P} a_x x \end{array}$.
  Then ϕ is R-linear and surjective, so there exists a R-linear map $\psi : P \to M$ s.t. $\phi\psi = id_P$. So $M \cong P \oplus \ker\phi$ and we have found a free R-module M s.t. P is a direct summand of M.

Theorem A.1.3. *Let R be a ring. TFAE*

1. *R is a semisimple ring.*
2. *all R-modules are projective*
3. *all finitely generated R-modules are projective*
4. *all cyclic R-modules are projective*

pf.

- $1 \leftrightarrow 2$ by the definition
- $2 \to 3 \to 4$ is obvious
- $4 \to 1$

  suppose all cyclic R-modules are projective. Let $I \subset R$ be a left ideal, then R/I is a cyclic R-module, hence projective. So there exists a s.e.s. $0 \to I \to R \to R/I \to 0$ which splits. So R is semisimple.

Theorem A.1.4 (Schur's Lemma)**.** *Let R be a ring, M, N be R-modules and $u : M \to N$ be a R-linear map.*

1. *if M is simple, then $u = 0$ or u is injective*
2. *if N is simple, then $u = 0$ or u is surjective*

pf. since $\ker u \subseteq M$, given M being simple, $\ker u$ is either M or 0 and the first part of the theorem follows. Similarly, $\mathrm{Im} u \subseteq N$, given N being simple, Im u is either N or 0 and the last part follows.

Definition A.1.11. *Let R be a ring. Write $S(R)$ for the set of isomorphism classes of simple R-modules.*

Lemma A.1.3. *Let M be a R-module and suppose there exists a simple R-module S and a set I and an isomorphism $\phi : M \to \bigoplus_{i \in I} S$. Then every simple R-submodule of M is isomorphic to S.*

pf. Let N be a simple R-submodule of M and suppose N is not isomorphic to S. Then $\forall i \in I$, the composition of the projection on the ith summand $\bigoplus S \to S$ and of ϕ is a R-linear map $N \to S$ which has to be zero by Schur's lemma. Then ϕ is zero on N, which implies $N = 0$, a contradiction to the fact that N is simple.

Theorem A.1.5. *Let R be a ring and M be a semisimple R-module. $\forall S \in S(R)$, let M_S be the sum of all the submodules of M that are isomorphic to S. Then* $\begin{cases} M = \bigoplus_{S\in S(R)} M_S \\ \exists I_S \text{ s.t. } M_S \cong \bigoplus_{i\in I_S} S, \ \ \forall S \in S(R) \end{cases}$. *Let N be a submodule of M. If we write $N \cong \bigoplus_{S\in S(R)} N_S$ and $N_S \cong \bigoplus_{S\in J_S} S$ as in the previous statements, then $\forall S \in S(R)$, $N_S = M_S \cap N$, and we can find an injection $J_S \hookrightarrow I_S$.*

pf.

- $\sum_{S\in S(R)} M_S$ is the sum of all the simple submodules of M. As M is semisimple, $M = \sum_{S\in S(R)} M_S$. Let $S \in S(R)$ and $X = S(R) \setminus \{S\}$. Also, let $N \equiv M_S \cap (\sum_{S'\in X} M_{S'})$. As N is a submodule of M, it is semisimple. So if it is nonzero, it is a simple submodule of M_S, hence isomorphic to S and also a simple submodule of $\sum_{S'\in X} M_{S'}$, hence isomorphic to an element of X, a contradiction. So $N = 0$ and the stmt follows.

- As M_S is a submodule of the semisimple R-module M, it is semisimple. Then M_S is the direct sum of a family of simple submodules by the theorem A.1.1. But every simple submodules of M_S is isomorphic to S.

- Let $S \in S(R)$. Then N_S is a sum of simple R-modules isomorphic to S, so $N_S \subset M_S$. On the other hand, $N \cap M_S$ is a direct sum of simple submodules of N and all these simple modules have to be isomorphic to S, so $N \cap M_S \subset N_S$. So we may assume $\begin{cases} M = M_S \\ N = N_S \end{cases}$ for some $S \in S(R)$. Then we can reduce the case into the one s.t. if S is simple R-module and I, J are two sets s.t. we have an injective R-linear map $u : N \equiv \bigoplus_{j\in J} S \hookrightarrow M \equiv \bigoplus_{i\in I} S$, then there exists an injection $J \hookrightarrow I$.
  For any R-modules M_1, M_2, let $\hom(M_1, M_2) \subset \hom_R(M_1, M_2)$ be the subgroup of R-linear maps that are 0 outside a finite length submodule of M_1. Then $\begin{cases} \hom(N, S) = \bigoplus_{j\in J} \mathrm{End}_R(S) \subset \hom_R(N, S) = \Pi_{j\in J}\mathrm{End}_R(S) \\ \hom(M, S) = \bigoplus_{i\in I} \mathrm{End}_R(S) \subset \hom_R(M, S) = \Pi_{i\in I}\mathrm{End}_R(S) \end{cases}$.
  Here $\mathbb{D} \equiv \mathrm{End}_R(S)$ is a division ring by Schur's lemma, and $\hom_R(M, S)$, $\hom_R(N, S)$ are naturally $\mathbb{D}$-modules and $\hom(M, S), \hom(N, S)$ are $\mathbb{D}$-submodules. Moreover, $\exists \begin{array}{rcl} \phi : \hom(M, S) & \to & \hom(N, S) \\ f & \mapsto & f \circ u \end{array}$ is $\mathbb{D}$-linear. Let $g \in \hom(N, S)$. As M is a semisimple R-module, $\exists N'$ a R-submodule of M s.t. $M = u(N) \oplus N'$. Define $f : M \to S$ by $f(u(x) + y) = g(x)$ if $x \in N$, $y \in N'$, since u is injective, and clearly R-linear. So $\phi(f) = g$ so it is surjective. Then the stmt follows by the incomplete basis theorem.

$\to$ the nonzero M_S are called the **isotypic components of** M and $M = \bigoplus_{S\in S(R)} M_S$ is called **the isotypic decomposition of** M**.**

Definition A.1.12. *A ring R is called simple if $R \neq 0$ and if the only ideals of R are 0 and R.*

Proposition A.1.1. *If R is a ring and $n \geq 1$ is an integer, then any ideal I of $M_n(R)$ is of the form $I = M_n(J)$, where J is a uniquely determined ideal of R.*

pf. If J is an ideal of R, $M_n(J)$ is an ideal of $M_n(R)$. Also if J, J' are two ideals of R s.t. $M_n(J) = M_n(J')$, then $J = J'$. So we just need to prove any ideal of $M_n(R)$ is of the form $M_n(J)$. Let I be an ideal of $M_n(R)$ and let J be the set of $(1,1)$-entries of elements of I. Let $x, y \in J$ and let $a \in R$. Choose matrices $X, Y \in I$ s.t. $(1,1)$-entries of X, Y are x, y respectively. Then $aX, Xa, X+Y$ are in I and their respective $(1,1)$-entries are $ax, xa, x+y$ so $ax, xa, x+y \in J$. So J is an ideal of R. Let's denote by E_{ij} the elementary matrices in $M_n(R)$. If $X = (x_{ij}) \in M_n(R)$, then $E_{ij}XE_{kl} = x_{jk}E_{il}$. If $X \in I$, $\forall j, k \in \{1, \ldots, n\}$, $E_{1j}XE_{k1} = x_{jk}E_{11} \in I$ so $x_{jk} \in J$ so $X \in M_n(J)$. This implies $I \subset M_n(J)$. Let $X = (x_{ij}) \in M_n(J)$. Then $X = \sum_{ij} x_{ij}E_{ij}$. Fix $i, j \in \{1, \ldots, n\}$. Choose $Y \in I$ s.t. $(1,1)$-entry of Y is x_{ij}. Then $E_{i1}YE_{1j} = x_{11}E_{ij} \in I$ so $M_n(J) \subset I$. Thus the proposition follows.

Corollary A.1.1. *If R is a simple ring, then so is $M_n(R)$, $\forall n \geq 1$.*

Theorem A.1.6. *Let $\mathbb{D}$ be a division ring, $n \geq 1$ be an integer and $R = M_n(\mathbb{D})$. Let $V = \mathbb{D}^n$. Then V is viewed as a left R-module by considering $\mathbb{D}^n$ as a space of $n \times 1$ matrices, and also viewed as the right $\mathbb{D}$-module $\mathbb{D}^n_{\mathbb{D}}$. Then*

1. *the ring R is simple, semisimple, left Artinian and left Noetherian.*
2. *R has a unique simple left module which is V. As a left R-module, R and V^n are isomorphic.*
3. *$End_{\mathbb{D}}(V) = R$*
4. *$End_R(V) = \mathbb{D}$*

pf.

- By the previous proposition, R is simple because $\mathbb{D}$ is. As a left $\mathbb{D}$-vector space, R is finite dimensional. Since left ideals of R are $\mathbb{D}$-vector subspaces of R, R is left Artinian and left Noetherian. Take a nonzero R-submodule W of V. Let $w \in W \setminus \{0\}$ and write $w = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$. Choose $i_0 \in \{1, \ldots, n\}$ s.t. $w_{i_0} \neq 0$. Then $\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = (w_{i_0}^{-1}E_{1i_0})w \in W$. If $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ is any element of

V, then $v = \begin{bmatrix} v_1 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ v_n & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in W$ so $W = V$. As the left R-module, $R \cong V^n$, the ring R is semisimple.

- Let M be a simple R-module. Choose $x \in M \setminus \{0\}$. Then the map $\begin{array}{rcl} u : R & \to & M \\ a & \mapsto & ax \end{array}$ is surjective, so M is isomorphic to a quotient of the R-module R. As $R \cong V^n$, $\;M \cong V$.

- Consider the map $\begin{array}{rcl} \phi : R & \to & \mathrm{End}_{\mathbb{D}}(V_{\mathbb{D}}) \\ a & \mapsto & (x \mapsto ax) \end{array}$. This map is well-defined because for all $a \in R$, $\forall x \in V$, $\lambda \in \mathbb{D}$, $\phi(a)(x\lambda) = a(x\lambda) = (ax)\lambda = (\phi(a)(x))\lambda$, so $\phi(a)$ is $\mathbb{D}$-linear. Let $a = (a_{ij}) \in R = M_n(\mathbb{D})$ s.t. $\phi(a) = 0$. Then $\forall j \in \{1, \dots, n\}$, if $e_j \in V$ is the element with jth entry equal to 1 and all other entries equal to 0, we have $0 = \phi(a)(e_j) = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}$ so $a = 0$, which implies ϕ is injective. Let $u \in \mathrm{End}_{\mathbb{D}}(V_{\mathbb{D}})$. $\forall j \in \{1, \dots, n\}$ if $e_j \in V$ is defined as above, $u(e_j) = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{bmatrix}$. Let $a = (a_{ij}) \in M_n(\mathbb{D})$. Then $\phi(a)(e_j) = u(e_j)$, $\forall j \in \{1, \dots, n\}$. As $(e_1, \dots, e_n)$ is a basis of $V_{\mathbb{D}}$ over $\mathbb{D}$, this implies $\phi(a) = u$ so ϕ is surjective.

- Let $E = \mathrm{End}_R(V)$. Write the action of the ring E on V on the right, that is, write $x(v) = vx$, $\forall v \in V$, $x \in E$. Let $\begin{array}{rcl} \psi : \mathbb{D} & \to & E \\ \lambda & \mapsto & (v \mapsto v\lambda) \end{array}$. As for λ, this map is well-defined, i.e. $\psi(\lambda)$ is R-linear $\forall \lambda \in \mathbb{D}$ and it is a morphism of rings. Because $\mathbb{D}$ is a division algebra, ψ is injective. Let $x \in E$ and define $\lambda \in \mathbb{D}$ by $e_1 x = \lambda e_1 + \sum_{j=2}^{n} \mu_j e_j$ s.t. $\mu_j \in \mathbb{D}$, $\;e_1, \dots, e_n \in V$ are as in the proof above. Let $v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in V$ and let $a = \begin{bmatrix} v_1 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ v_n & 0 & \dots & 0 \end{bmatrix} \in M_n(\mathbb{D}) = R$. Then $v = ae_1$ so $vx = (ae_1)x = a(e_1 x) = a \begin{bmatrix} \lambda \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} = \begin{bmatrix} v_1\lambda \\ \vdots \\ v_n\lambda \end{bmatrix} = v\lambda$ so $x = \psi(\lambda)$ and ψ is surjective.

Corollary A.1.2. *If $\mathbb{D}, \mathbb{D}'$ are two division rings and $n, n' \geq 1$ are two integers s.t. $M_n(\mathbb{D}) \cong M_{n'}(\mathbb{D}')$ as rings, then $\mathbb{D} \cong \mathbb{D}'$ and $n = n'$.*

pf. Let $R = M_n(\mathbb{D}) \cong M_{n'}(\mathbb{D}')$ and let M be the unique simple R-module by the previous theorem 2. Then by the previous theorem 3, $\mathbb{D}' \cong \mathrm{End}_R(M) \cong \mathbb{D}$. Hence $n = \dim_{\mathbb{D}}(M) = \dim_{\mathbb{D}'}(M) = n'$.

Definition A.1.13. *If R is a ring, we denote by R^{op} its opposite ring; it is isomorphic to R as an additive group and its multiplication is given by $ab(R^{op}) = ba(R)$.*

Theorem A.1.7 (Double Centraliser Property)**.** *Let R be a simple ring and I be a nonzero left ideal of R. Let $D = End_R(I)$ and make I a right D^{op}-module by setting $xu = u(x)$ if $\left\{ \begin{array}{c} x \in I \\ u \in D \end{array} \right.$. Then the map $\begin{array}{rcl} f : R & \to & End_{D^{op}}(I) \\ a & \mapsto & (x \mapsto ax) \end{array}$ is an isomorphism of rings.*

pf. As $I \neq 0$, $\ker f \neq R$. As $\ker f$ is an ideal of R and R is simple, $\ker f = 0$, i.e. f is injective. Let $E = \mathrm{End}_{D^{op}}(I)$. Make I a left E-module by setting $hx = h(x)$, $\forall x \in I$, $h \in E$. Then $\forall x \in I$, $\forall h \in E$, $hf(x) = f(hx)$. Indeed if $a \in I$, $\begin{array}{rcl} r_a : I & \to & I \\ y & \mapsto & ya \end{array}$ is in D so $h(xa) = h(r_a(x)) = h(xr_a) = h(x)r_a = h(x)a$ so $(hf(x))(a) = h(xa) = h(x)a = f(h(x))(a)$. Thus $Ef(I) \subset f(I)$. Since I is a nonzero left ideal of R, IR is a nonzero ideal of R, hence $IR = R$ as R is simple. Then $f(I)f(R) = f(R)$ and $E = Ef(R) = Ef(I)f(R) \subset f(I)f(R) = f(R)$, so f is surjective.

Corollary A.1.3. *If R is a simple ring with a minimal nonzero left ideal, then there exists a unique division ring $\mathbb{D}$ and a unique integer $n \geq 1$ s.t. $R \cong M_n(\mathbb{D})$. In particular, a simple ring is left Artinian iff it is of the form $M_n(\mathbb{D})$ with $\mathbb{D}$ a division ring and $n \in \mathbb{Z}_{++}$.*

pf. Let $I \subset R$ be a minimal nonzero left ideal. Then I is simple R-module, so $\mathbb{D} \equiv \mathrm{End}_R(I)$ is a division ring by Schur's lemma. Make I a right $\mathbb{D}^{op}$-module as in the previous theorem. Then $R \cong \mathrm{End}_{\mathbb{D}^{op}}(I)$. Let $E = \mathrm{End}_{\mathbb{D}^{op}}(I)$ and let $E_f = \{u \in E | rk(u) < \infty\}$ where $rk(u)$ is the dimension over $\mathbb{D}^{op}$ of the image of u. Then $E_f \neq 0$ and E_f is an ideal of E. Since $E \cong R$ is a simple ring, $E_f = E$, hence the identity of I is in E_f and I is a finite dimensional right $\mathbb{D}^{op}$-vector space. The uniqueness follows by the previous corollary.

Proposition A.1.2. *Let $R_1, \ldots, R_n$ be rings and let $R = R_1 \times \cdots \times R_n$. Then every R-module is of the form $M = M_1 \times \cdots \times M_n$ with M_i a R_i-module for $1 \leq i \leq n$.*

pf. In R, write $1 = e_1 + \cdots + e_n, \quad \forall e_i \in R_i$. As an element of $R_1 \times \cdots \times R_n, \quad e_i$ is the n-tuple with ith entry equal to $1 \in R_i$ and 0 for all other entries. Then all e_i are central in R. Let M be a R-module, set $M_i = e_i M$. Then R acts on M_i through the obvious projection $R \to R_i$, so we need to show $M \cong M_1 \times \cdots \times M_n$. Consider the map $\begin{array}{ccc} M_1 \times \cdots \times M_n & \to & M \\ (x_1, \dots, x_n) & \mapsto & \sum_i x_i \end{array}$ which is the R-linear map. If $x \in M, \quad x = xe_1 + \cdots + xe_n$ with $xe_i \in M_i, \quad \forall i$ so u is surjective. If $\sum x_i = 0, \quad x_i \in M_i$ then $\forall j \in \{1, \dots, n\}, \quad 0 = e_j(x_1 + \cdots + x_n) = e_j x_j = x_j$ so u is injective.

Corollary A.1.4. *Suppose $R = R_1 \times \cdots \times R_n$ as in the previous proposition.*

1. *R is semisimple iff all the R_i are semisimple*

2. *Every simple R-module is of the form $0 \times \cdots \times 0 \times M_i \times 0 \times \cdots \times 0, \quad i \in \{1, \dots, n\}$ and M_i a simple R_i-module*

Corollary A.1.5. *Let $\mathbb{D}_1, \dots, \mathbb{D}_r$ be division rings, and $n_1, \dots, n_r \geq 1$ be integers. Then $R \equiv M_{n_1}(\mathbb{D}_1) \times \dots M_{n_r}(\mathbb{D}_r)$ is a semisimple ring and its simple modules are $\mathbb{D}_1^{n_1}, \dots, \mathbb{D}_r^{n_r}$.*

Let R be a ring and I be a left ideal of R. Denote $\mathcal{I}_I \subset R$ the sum of all the left ideals I' of R that are isomorphic to I as R-modules. Then it is a left ideal of R.

Lemma A.1.4. *Let R be a ring and I be a minimal nonzero left ideal of R. Then $\mathcal{I}_I$ is an ideal of R. Moreover if I, J are two nonisomorphic minimal nonzero left ideals of R, then $\mathcal{I}_I \mathcal{I}_J = 0$.*

pf. Let I' be a left ideal of R s.t. $I' \cong I$ as R-modules, and let $a \in R$. Then we have a surjective R-linear map $\begin{array}{ccc} I' & \to & I'a \\ x & \mapsto & xa \end{array}$ and I' is a simple R-module, so $I'a = 0$ or $I'a \cong I'$. If $I'a = 0, \quad I'a \subset \mathcal{I}_I$, otherwise $I'a$ is another left ideal of R isomorphic to I, so $I'a \subset \mathcal{I}_I$. thus $\mathcal{I}_I$ is a right ideal.
Let J be another minimal nonzero left ideal of R, and suppose $\mathcal{I}_I \mathcal{I}_J \neq 0$. Then there are left ideals I', J' of R and an element $a \in J'$ s.t. $I' \cong I, \quad J' \cong J, \quad I'a \neq 0$. As J' is a simple R-module and $I'a \subset J'$ is a nonzero submodule of $J', \quad I'a = J'$. As I' is a simple R module, the surjective map $\begin{array}{ccc} I' & \to & I'a \\ x & \mapsto & xa \end{array}$ is an isomorphism. So $I \cong I' \cong I'a = J' \cong J$.

Theorem A.1.8 (Artin-Wedderburn). *Let R be a semisimple ring. Let $(I_i)_{i \in A}$ be a set of representatives of the isomorphism classes of minimal nonzero left ideals of R, and let $R_i = \mathcal{I}_{I_i} \subset R$. Then A is finite so we choose an identification $A = \{1, \dots, r\}$. Moreover, all the R_i are rings with units, $R \cong R_1 \times \cdots \times R_r$ as rings and $\forall i \in \{1, \dots, r\}$, there exists a unique division ring $\mathbb{D}_i$ and a unique integer $n_i \geq 1$ s.t. $R_i \cong M_{n_i}(\mathbb{D}_i)$.*

pf. As R is a semisimple ring, R is a direct sum of simple submodules by the theorem A.1.1. So $R = \bigoplus_{i \in A} R_i$ by the theorem A.1.5. Then A is finite.
By the previous lemma, every $\mathcal{I}_i$ is an ideal of R. As $R = \bigoplus_{i=1}^r R_i$, all the R_i are rings with units and $R \cong R_1 \times \cdots \times R_r$ as rings. Fix $i \in \{1, \ldots, r\}$. Let $J \neq 0$ be an ideal of R_i. As J is also an ideal of R, it contains a minimal nonzero left ideal I of R. By definition of $I_1, \ldots, I_r, \ , \ \exists j \in \{1, \ldots, r\}$ s.t. $I \cong I_j$ as R-modules. As $I \subset J \subset R_i, \ j = i$. So $R_i = \mathcal{I}_{I_i} = \mathcal{I}_I$. Hence it suffices to show if I' is a left ideal of R s.t. $I \cong I'$, then $I' \subset J$. Fix such a I' and let $\phi : I \to I'$ be an isomorphism of R-modules. As R is a semisimple ring, there exists a left ideal I'' of R s.t. $R = I \oplus I''$. Write $1 = e + e'', \ e \in I, \ e'' \in I''$. Then $e^2 = e, \ I = Re$. So $I' = \phi(I) = \phi(Re) = \phi(Re^2) = \phi((Re)e) = \phi(Ie) = I\phi(e) \subset J$.

e.g. let K be a field and suppose R is a semisimple K-algebra that is finite dimensional as a K-vector space. Then its simple factors R_i are also K-algebras and so are the division rings $\mathbb{D}_i$ with $\dim_K \mathbb{D}_i < \infty$. In particular, if K is algebraically closed, all the $\mathbb{D}_i$ are equal to K, so $R \cong M_{n_1}(K) \times \cdots \times M_{n_r}(K)$.

Definition A.1.14. *Let R be a ring. The Jacobson radical of R is the intersection of all the maximal left ideals of R, denoted by $rad(R)$.*

A.1.2 The Representation Ring

Definition A.1.15. *Let R be a ring. Define $K(R)$ to be the quotient of the free abelian group on the basis elements $[M]$, for M a finite length R-module, by all the relations of the form $[M] = [M'] + [M'']$, where $0 \to M' \to M \to M'' \to 0$ is an exact sequence of R-modules. It is called the Grothendieck group of the category of finite length R-modules. Define $PK(R)$ to be the quotient of the free abelian group on the basis elements $[P]$, for P a finite length projective R-module, by all the relations of the from $[P] = [P'] + [P'']$ where $0 \to P' \to P \to P'' \to 0$ is an exact sequence of R-modules. It is called the Grothendieck group of the category of finite length projective R-modules.*

$\to 0 \to 0 \to 0 \to 0 \to 0$ gives $[0] + [0] = [0]$ in $K(R), PK(R)$ so $[0] = 0$. Then for an isomorphism of R-modules $M \to M', \ 0 \to M \to M' \to 0$ gives $[M] = [M'] + [0] = [M']$ in $K(R)$, and similarly for $PK(R)$. So if $0 = M_n \subset M_{n-1} \subset \cdots \subset M_0 = M, \ , \ [M] = \sum_{i=1}^n [M_{i-1}/M_i]$ in $K(R)$.

Proposition A.1.3. *As a group, $K(R)$ is the free abelian group with basis $\{[M] : M \in S(R)\}$, where $S(R)$ is the set of isomorphism classes of simple R-modules.*

pf. Let S be the free abelian group on the set $S(R)$, and denoted by $(e_M)_{M \in S(R)}$ its canonical basis. Define $\begin{array}{rcl} \phi : S & \to & K(R) \\ e_M & \mapsto & [M] \end{array}$, $\begin{array}{rcl} \psi : K(R) & \to & S \\ M & \mapsto & \sum e_{M_{i-1}/M_i} \end{array}$ where

$M = M_0 \supset M_1 \supset \cdots \supset M_n = 0$. If $0 \to M' \to M \to_u M'' \to 0$ is a s.e.s. of finite length R-modules, choose Jordan-Hölder series $\begin{cases} M' = M'_0 \supset \cdots \supset M'_n = 0 \\ M'' = M''_0 \supset \cdots \supset M''_m = 0 \end{cases}$. $\forall 0 \leq j \leq m,\ M_j = u^{-1}(M''_j)$. Then $M = M_0 \supset \cdots \supset M_m = M'_0 \supset \cdots \supset M'_n = 0$ is a Jordan-Hölder series for M, which gives the desired equality. So ψ is well-defined and ψ, ϕ are inverses of each other, so the proposition follows.

Proposition A.1.4. *If R is a semisimple ring, the $K(R) = PK(R)$ and for all finite length R-modules M, M', $[M] = [M']$ in $K(R)$ iff $M \cong M'$.*

pf. If R is semisimple, every R-module is projective, so $K(R) = PK(R)$. Let M, M' be two R-modules of finite length. As R is semisimple, $\begin{cases} M \cong \bigoplus_{N \in S(R)} N^{\oplus r_N} \\ M' \cong \bigoplus_{N \in S(R)} N^{\oplus r'_N} \end{cases}$. Then $[M] = \sum r_N[N],\ [M'] = \sum r'_N[N]$. Then by the previous proposition, $[M] = [M']$ iff $r_N = r'_N,\ \forall N \in S(R)$, which is equivalent to $M \cong M'$.

Definition A.1.16. *The representation ring of G over k, denoted by $R_k(G)$ is defined as $R_k(G) = K(k[G])$. Also $P_k(G) = PK(k[G]),\ S_k(G) = S(k[G])$.*

$\to$ $\forall [M], [M'] \in R_k(G),\ [M][M'] = [M \otimes_k M']$ defines a biadditive map $R_k(G) \otimes R_k(G) \to R_k(G)$ with unit element equal to $[\mathbb{1}_k]$. By the properties of tensor products, it is commutative.

Proposition A.1.5. *If M is a $k[G]$-module and N is a projective $k[G]$-module, then $k[G]$-module $M \otimes_k N$ is also projective.*

pf. As N is projective over $k[G]$, it is a direct summand of some free $k[G]$-module $k[G]^{\oplus I}$ and then $M \otimes_k N$ is a direct summand of $M \otimes_k k[G]^{\oplus I} = (M \otimes_k k[G])^{\oplus I}$. So we can reduce into the case where $M \otimes_k k[G]$. Let $\underline{M}$ be the k-vector space M, considered as a representation of G with trivial action. Define a k-linear map $\begin{array}{rcl} u : \underline{M} \otimes_k k[G] & \to & M \otimes_k k[G] \\ x \otimes g & \mapsto & gx \otimes g \end{array}$. Then k-linear map $\begin{array}{rcl} v : M \otimes_k k[G] & \to & \underline{M} \otimes_k k[G] \\ x \otimes g & \mapsto & g^{-1}x \otimes g \end{array}$ is an inverse of u, so u is an isomorphism of k-modules. Also $\forall g, h \in G,\ \ \forall x \in \underline{M}$, $\begin{cases} h \cdot u(x \otimes g) = h \cdot (gx \otimes g) = (hgx) \otimes (hg) \\ u(h \cdot (x \otimes g)) = u(x \otimes (hg)) = (hgx) \otimes (hg) \end{cases}$ gives u is G-equivariant. Since $\underline{M} \otimes_k k[G]$ is a direct sum of copies of $k[G]$, the $k[G]$-module $M \otimes_k k[G]$ is free, and the proposition follows.

e.g. $G = \mathbb{Z}/n\mathbb{Z}$, k a field containing all the nth root of unity, and char$(k) \nmid n$. Then $R_k(G) \cong \mathbb{Z}^n$ as a group and $R_k(G)$ is isomorphic to the group algebra $\mathbb{Z}[\mathbb{Z}/n\mathbb{Z}]$ as a ring.

A.1.3 Induction and Frobenius Reciprocity

Let R be a ring and $\alpha : H \to G$ be a morphism of groups. Then α gives a morphism of rings $\phi : R[H] \to R[G]$.

Definition A.1.17. *If M is a $R[G]$-module, we can see it as a $R[H]$-module by making $R[H]$ act via the morphism $\phi : R[H] \to R[G]$. The resulting $R[H]$-module is denoted by $Res_H^G M$ and called the restriction of M to H.*

$\to$if $u : M \to N$ is a morphism of $R[G]$-modules, $\mathrm{Res}_H^G(u) : \mathrm{Res}_H^G M \to \mathrm{Res}_H^G N$ for the same map u, seen as a morphism of $R[H]$-modules.

Definition A.1.18. *If M is a $R[H]$-module, set $Ind_H^G M = R[G] \otimes_{R[H]} M$ where $R[H]$ acts on the left on $R[G]$ via the morphism ϕ. This is called the induction of M from H to G.*

$\to$ if $u : M \to N$ is a morphism of $R[H]$-modules, $\mathrm{Ind}_H^G(u) : \mathrm{Ind}_H^G M \to \mathrm{Ind}_H^G N$ for the $R[G]$-linear map $id \otimes u$.

Definition A.1.19. *If M is a $R[H]$-module, we set $CoInd_H^G M = \hom_{R[H]}(R[G], M)$ where $R[G]$ is seen as a left $R[H]$-module via $\phi : R[H] \to R[G]$. We make this into a left $R[G]$-module using the right regular action of $R[G]$ on itself, i.e. $\forall x \in R[G],\ u \in CoInd_H^G M,\ (x \cdot u)(y) = u(yx),\ \forall y \in R[H]$. The $R[G]$-module $CoInd_H^G$ is called the coinduction of M from H to G.*

$\to$ if $u : M \to N$ is a morphism of $R[H]$-modules, we write $\mathrm{CoInd}_H^G(u)$: $\mathrm{CoInd}_H^G M \to \mathrm{CoInd}_H^G N$ for the $R[G]$-linear map sending $v \in \hom_{R[H]}(R[G], M)$ to $u \circ v$.

Let M be a $R[H]$-module. Then restriction map $R[G] \to M$ along the inclusion $G \subset R[G]$ induces an isomorphism of R-modules

$$\psi : \mathrm{CoInd}_H^G M \to \{f : G \to M \mid \forall h \in H,\ \forall g \in G,\ f(\alpha(h)g) = hf(g)\}$$

pf. Because G generates the $R[H]$-module $R[G]$, so a $R[H]$-linear map $u : R[G] \to M$ is uniquely determined by its restriction to G. So ψ is injective. Let $f : G \to M$ satisfying the condition in the formula above, and define a R-linear map $\begin{array}{ccc} R[G] & \to & M \\ \sum_{g\in G} c_g g & \mapsto & \sum c_g f(g) \end{array}$. Then $\forall h \in H,\ x = \sum_{g\in G} c_g c \in R[G],\ u(hx) = u(\sum c_g \alpha(h) g) = \sum c_g f(\alpha(h)g) = \sum c_g h f(g) = hu(x)$, i.e. u is $R[H]$-linear. Then $\psi(u) = f$, so ψ is surjective.

Proposition A.1.6. *If $\alpha : H \to G$ is injective, then $\phi : R[H] \to R[G]$ makes $R[G]$ into a free $R[H]$-module. More precisely, let $(g_i)_{i\in I}$ be a complete set of representatives of $\alpha(H) \subset G$ in G, then $(g_i)_{i\in I}$ is a basis of $R[G]$ as a left/right $R[H]$-module.*

Definition A.1.20. *Suppose $G=\{1\}$. Then for every $R[H]$-module M, $\mathrm{Ind}_H^1 M$ is called the R-module of coinvariants of M under H and denote by M_H.*

$\rightarrow M_H = M/(hm-m)$ the quotient of M by the R-submodule generated by all $hm-m$.

Lemma A.1.5. *Let M be a $R[H]$-module.*

1. *if $\alpha : H \to G$ is surjective, then $\mathrm{Ind}_H^G M = M_{\ker\alpha}$, with the following action of G; $\forall g \in G, \quad x \in M_{\ker\alpha}$, choose preimage $h \in H$ of g by α and $m \in M$ of x by the obvious quotient map, and then gx is the image in $M_{\ker\alpha}$ of hm.*
2. *$\mathrm{Ind}_H^G M = \mathrm{Ind}_{\alpha(H)}^G M_{\ker\alpha}$*

pf.

- Let $u : \mathrm{Ind}_H^G M = R[G] \otimes_{R[H]} M \to M_{\ker\alpha}$ be defined as follows; $\forall a \in R[G], \ m \in M$, choose $b \in R[H]$ s.t. $\phi(b) = a$ and take for $u(a \otimes m)$ the image of bm in $M_{\ker\alpha}$. Because $\ker\phi$ is the $R[H]$-submodule of $R[H]$, generated by $h-1, \ h \in \ker\alpha$ this does not depend on the choice of h. Since the formula $u(a\otimes m)$ is additive in a and m and takes the same value on $(a\phi(x), m)$ and (a, xm) if $x \in R[H]$, the map is well defined on $R[G]\otimes_{R[H]} M$. $\begin{array}{rcl} v' : M & \to & R[G]\otimes_{R[H]} M \\ m & \mapsto & 1 \otimes m \end{array}$. If $m \in M, \ h \in \ker\alpha, \ v'(hm - m) = 1\otimes hm - 1 \otimes m = \alpha(h)\otimes m - 1 \otimes m = 0$ so v' defines a R-linear map $v : M_{\ker\alpha} \to \mathrm{Ind}_H^G M$. Here u and v' are inverses to each other, so the statement follows.
- $$\begin{aligned} \mathrm{Ind}_H^G M &= \mathrm{Ind}_{\alpha(H)}^G \mathrm{Ind}_H^{\alpha(H)} M && \text{transitivity of } \otimes \\ &= \mathrm{Ind}_{\alpha(H)}^G M_{\ker\alpha} && \text{statement 1} \end{aligned}$$

Theorem A.1.9. *For every exact sequence $0 \to M' \to M \to M'' \to 0$ of $R[H]$-modules, the sequence*

$$\mathrm{Ind}_H^G M' \to \mathrm{Ind}_H^G M \to \mathrm{Ind}_H^G M'' \to 0$$

is exact. Moreover $0 \to \mathrm{Ind}_H^G M' \to \mathrm{Ind}_H^G M \to \mathrm{Ind}_H^G M'' \to 0$ is exact if
$$\begin{cases} \alpha : H \to G \text{ is injective} \\ \ker\alpha < \infty, \ R \text{ is semisimple, } |\ker\alpha| \text{ is invertible in } R \end{cases}$$

pf. By the property of tensor product the first stmt follows.
The right $R[H]$-module $R[G]$ is free by the previous proposition and therefore the tensor products by $R[G]$ over $R[H]$ preserves exact sequence.
By the previous lemma, $\mathrm{Ind}_H^G M = \mathrm{Ind}_{\alpha(H)}^G M_{\ker\alpha}$ for every $R[H]$-module M. Let

$f : M \to N$ be an injective $R[G]$-linear map. As $R[G]$ is semisimple, there exists a $R[G]$-module M' of N s.t. $N = M \oplus M'$. Then $N_G = M_G \oplus M'_G$ and $M_G \to N_G$ is injective. So coinvariants preserves injection and therefore we have the exact sequence preserved.

Definition A.1.21. *Suppose $G = \{1\}$. Then for every $R[H]$-module M, $CoInd_H^1 M$ is called the R-module invariants of M under H and denoted by M^H.*

$\to M^H = \{m \in M | \forall h \in H, \ \ hm = m\}$.

Lemma A.1.6. *Let M be a $R[H]$-module.*

1. *if $\alpha : H \to G$ is surjective, $CoInd_H^G M = M^{\ker\alpha}$ with the following action of G; $\forall g \in G$, $\forall m \in M^{\ker\alpha}$, choose a preimage $h \in H$ of g by α and then gm is defined to be hm.*
2. *$CoInd_H^G M = CoInd_{\alpha(H)}^G M^{\ker\alpha}$*

pf.

- Let $u : \mathrm{CoInd}_H^G M = \hom_{R[H]}(R[G], M) \to M^{\ker\alpha}$ be defined as follows; if $f : R[G] \to M$ is a $R[H]$-liner map, take $u(f) = f(1)$. Then it is well-defined because $\forall h \in \ker\alpha$, $hf(1) = f(\alpha(h)1) = f(1)$. Let $v : M^{\ker\alpha} \to \hom_{R[H]}(R[G], M)$ be the map defined follows; if $m \in M$, $a \in R[G]$, choose $b \in R[H]$ s.t. $\phi(b) = a$ and set $v(m)(a) = bm$. This does not depend on the choice of b because $\ker\phi$ is the $R[H]$-submodule of $R[H]$ generated by all $h - 1$, $h \in \ker\alpha$. Then u and v are inverses to each other and the stmt follows.
- $$\begin{aligned} \mathrm{CoInd}_H^G M &= \mathrm{CoInd}_{\alpha(H)}^G \mathrm{CoInd}_H^{\alpha(H)} M && \text{properties of homomorphisms} \\ &= \mathrm{CoInd}_{\alpha(H)}^G M^{\ker\alpha} && \text{by the statement 1} \end{aligned}.$$

Theorem A.1.10. *For every exact sequence $0 \to M' \to M \to M'' \to 0$ of $R[H]$-modules, the sequence $0 \to CoInd_H^G M' \to CoInd_H^G M \to CoInd_H^G M''$ is exact. Moreover $0 \to CoInd_H^G M' \to CoInd_H^G M \to CoInd_H^G M'' \to 0$ if*
$$\begin{cases} \alpha : H \to G \text{ is injective} \\ \ker\alpha < \infty, \ \ R \text{ is semisimple}, \ |\ker\alpha| \text{ is invertible in } R \end{cases}$$

pf. By the left-exactness of hom, the first stmt follows.
By the previous proposition, the left $R[H]$-module $R[G]$ is free, so taking $\hom_{R[H]}(R[G], \cdot)$ preserves exact sequences.
By the previous lemma, $\mathrm{CoInd}_H^G M = \mathrm{CoInd}_{\alpha(H)}^G M^{\ker\alpha}$ for every $R[H]$-module M.
Let $M \to N$ be a surjective $R[G]$-linear map. As $R[G]$ is semisimple, there exists a $R[G]$-submodule N of M s.t. $M = N \oplus N'$ and by the definition of invariants, $M^G = N^G \oplus N'^G$ and $M^G \to N^G$ is surjective. So invariants preserves surjection and therefore we get a s.e.s.

Definition A.1.22. *Let M be a $R[H]$-module. We denote by* $\begin{array}{rcl} \epsilon_M : \ Res_H^G CoInd_H^G M & \to & M \\ u & \mapsto & u(1) \end{array}$, *where* $u \in \hom_{R[H]}(R[G], M)$. *Additionally denote by* $\begin{array}{rcl} \eta_M : M & \to & Res_H^G Ind_H^G M \\ m & \mapsto & 1 \otimes m \end{array}$.

$\to \forall x \in R[H],\ \forall u \in \hom_{R[H]}(R[G], M),\ \ \epsilon_M(x \cdot u) = (x \cdot u)(1) = u(x) = xu(1)$ so it is $R[H]$-linear. η_M is clearly $R[H]$-linear.

Theorem A.1.11 (Frobenius Reciprocity)**.** *Let M be a $R[G]$-module and N be a $R[H]$-module.*

1. *the morphism of groups*

$$\begin{array}{rcl} \Phi : \hom_{R[G]}(M, CoInd_H^G N) & \to & \hom_{R[H]}(Res_H^G M, N) \\ u & \mapsto & \epsilon_N \circ Res_H^G(u) \end{array}$$

   is an isomorphism.

2. *the morphism of groups*

$$\begin{array}{rcl} \Phi' : \hom_{R[G]}(Ind_H^G N, M) & \to & \hom_{R[H]}(N, Res_H^G M) \\ u & \mapsto & Res_H^G(u) \circ \eta_N \end{array}$$

   is an isomorphism.

pf. Let $R_1 = R[H],\ \ R_2 = R[G]$ for simplicity.

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$$\begin{array}{rcl} \Psi : \hom_{R[H]}(\mathrm{Res}_H^G M, N) = \hom_{R_1}(M, N) & \to & \hom_{R[G]}(M, \mathrm{CoInd}_H^G N) = \hom_{R_2}(M, \hom_{R_1}(R \\ u : M \to N & \mapsto & m \mapsto (a \mapsto (u \end{array}$$

  Let u be a R_1-linear map from M to N. Then $\Psi(u) : M \to \hom_{R_1}(R_2, N)$ is the R_2-linear map $m \mapsto (a \mapsto (u(am)))$ and $\Phi\Psi(u) : M \to N$ is the R_1-linear map sending $m \in M$ to $(\Psi(u)(m))(1) = u(m)$. So $\Phi\Psi(u) = u$. Conversely, let v be a R_2-linear map from M to $\hom_{R_1}(R_2, N)$. Then $\Phi(v)$ is the R_1-linear map $m \mapsto v(m)(1)$ and $\forall m \in M,\ \ a \in R_2,\ \ (\Psi\Phi(v)(m))(a) = \Phi(v)(am) = (v(am))(1) = (a \cdot v(m))(1) = v(m)(a),\ \ \Psi\Phi(v) = v$. Thus they are inverses to each other, so isomorphism.

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$$\begin{array}{rcl} \Psi' : \hom_{R[H]}(N, \mathrm{Res}_H^G M) = \hom_{R_1}(N, M) & \to & \hom_{R[G]}(\mathrm{Ind}_H^G N, M) = \hom_{R_2}(R_2 \otimes_{R_1} N, M) \\ u : N \to M & \mapsto & a \otimes n \mapsto au(n) \end{array}$$

  Let u be a R_1-linear map from N to M. Then $\forall n \in N,\ \ \Phi'\Psi'(u)(n) = \Psi'(u)(1 \otimes n) = u(n),\ \ \Phi'\Psi'(u) = u$. Let v be a R_2-linear map from $R_2 \otimes_{R_1} N$ to M. Then $\forall a \in R_2,\ \ \forall n \in N,\ \ \ \Psi'\Phi'(v)(a \otimes n) = a\Phi'(v)(n) = av(1 \otimes n) = v(a \otimes n),\ \ \Psi'\Phi'(v) = v$. Thus they are inverses to each other, so isomorphism.

A.2 Finite Group Representation

A.2.1 Introduction

Let R be a ring and G be a group. A $R[G]$-module is a R-module M with a R-linear action of G, i.e. a morphism of monoids $\rho : G \to \mathrm{End}_R(M)$. This is called a R-linear representation of G on the R-module M, denoted by (M, ρ). The representation (M, ρ) is called **irreducible** if the $R[G]$-module M is simple and it is called **faithful** if ρ is injective. A $R[G]$-linear map is also called a G-equivariant map. A sub-$R[G]$-module is also called a **subrepresentation**. **The regular representation of** G is the representation corresponding to the left regular $R[G]$-module.

Let $\begin{array}{ccc} \epsilon : R[G] & \to & R \\ \sum_{g\in G} \alpha_g g & \mapsto & \sum_{g\in G} \alpha_g \end{array}$ be called the augmentation map. It is a surjective R-linear ring homomorphism, and its kernel is called the augmentation ideal of $R[G]$.

Theorem A.2.1. *Let R be a ring and G be a finite group. Then $R[G]$ is a semisimple ring iff R is a semisimple ring and $|G|$ is invertible in R.*

pf.

- Let M be a $R[G]$-module and let N be a $R[G]$-submodule of M. As R is semisimple, $\exists N'$, a R-submodule of M s.t. $M = N \oplus N'$, so there exists a surjective R-linear map $f : M \to N$ s.t. $f|_N = id_N$. Define $\begin{array}{ccc} F : M & \to & N \\ v & \mapsto & |G|^{-1} \sum_{g\in G} g^{-1} f(gv) \end{array}$. Then it is R-linear and $\forall g \in G,\ \forall v \in M,\ \ F(gv) = |G|^{-1} \sum_{h\in G} h^{-1} f(hgv) = |G|^{-1} g \sum_{h\in G} (hg)^{-1} f(hgv) = gF(v)$, i.e. $F(gv) = gF(v)$. Thus $F(v) = |G|^{-1} \sum g^{-1} f(gv) = |G|^{-1} \sum g^{-1} gv = v$, i.e. $F|_N = id_N$. Here $\forall v \in M,\ F(v) \in N$ and therefore $FF(v) = F(v)$, hence $F(v - F(v)) = 0,\ v = (v - F(v)) + F(v) \in \ker F + N$. If $v \in N \cap \ker F,\ \ v = F(v) = 0$. Thus $M = N \oplus \ker F$.

- Assume $R[G]$ is semisimple. Then there exists the augmentation map ϵ. It makes R into a $R[G]$-module, which is automatically semisimple by assumption. As $R[G]$ acts on R through its quotient $R[G]/\ker\epsilon \cong R$, the R-module R is semisimple, so R is a semisimple ring. It remains to show $|G|$ is invertible in R; by the remark below it suffices to show $|G| \neq 0$ in each $\mathbb{D}_i$ appearing in the Artin-Wedderburn decomposition of R. Suppose $|G| = 0$ in $\mathbb{D} \equiv \mathbb{D}_i$. As $M_{n_i}(\mathbb{D})[G] \cong M_{n_i}(\mathbb{D}[G])$ is a quotient of $R[G]$, it is semisimple, hence so is $\mathbb{D}[G]$ by the proposition on ideals of matrix rings. Let $c = \sum_{g\in G} g \in \mathbb{D}[G]$. Then $c \neq 0$, c is central, and $c^2 = |G|c = 0$, so the left ideal $I = \mathbb{D}[G]c$ satisfies $Ic = \mathbb{D}[G]c^2 = 0$. By semisimplicity write $\mathbb{D}[G] = I \oplus J$ and $1 = e + f$

with $e \in I,\ f \in J$. Then $c = ec + fc = cf \in J$, using $ec \in Ic = 0$ and the centrality of c, while $c = 1c \in I$, so $c \in I \cap J = 0$, a contradiction.

By the Artin-Wedderburn theorem, $R = M_{n_1}(\mathbb{D}_1) \times \cdots \times M_{n_r}(\mathbb{D}_r)$, where $\mathbb{D}_j$ are division rings and $n_j \geq 1,\ \forall j$. For every i, let $R_i = M_{n_i}(\mathbb{D}_i)$. Then we have a surjective morphism of rings $R[G] \to R_i$ and therefore R_i is also semisimple. Also $|G|$ is invertible in R iff it is invertible in every R_i, iff it is invertible in every $\mathbb{D}_i$, iff it is nonzero in $\mathbb{D}_i$.

Theorem A.2.2. *1. There are uniquely determined k-division algebras $\mathbb{D}_1, \ldots, \mathbb{D}_r$ and integers $n_1, \ldots, n_r \geq 1$ s.t. $\dim_k(\mathbb{D}_i)$ is finite for every i and*

$$k[G]/rad(k[G]) \cong M_{n_1}(\mathbb{D}_1) \times \cdots \times M_{n_r}(\mathbb{D}_r)$$

A complete set of representatives of the isomorphism classes of irreducible representations of G is given by $V_1 \equiv \mathbb{D}_1^{n_1}, \ldots, V_r \equiv \mathbb{D}_r^{n_r}$ and we have a G-equivariant isomorphism

$$k[G]/rad(k[G]) \cong V_1^{n_1} \oplus \cdots \oplus V_r^{n_r}$$

2. If k is algebraically closed, then $\mathbb{D}_i = k,\ \forall i$, and

$$\sum_V (\dim V)^2 = \sum_{i=1}^r \dim(V_i)^2 = \sum_{i=1}^r n_i^2 = \dim(k[G]/rad(k[G])) \leq \dim(k[G]) = |G|$$

where the first sum is taken over the isomorphism classes of irreducible representations of G. The inequality is an equality iff the characteristic of k does not divide $|G|$.

Definition A.2.1. *If R is a ring and G is a group, denote by $S_R(G)$ the set of isomorphism classes of irreducible representations of G on R-modules.*

e.g.

- Fix R, G as above. The trivial representation of G over R is the representation of G on R given by the augmentation map $\epsilon : R[G] \to R$ by the trivial action of G on R, denoted by $\mathbb{1}$.
- If G is the symmetric group $\mathcal{G}_n$ and R is any ring, we denote by *sgn* the representation of G on R given by the sign morphism $G \to \{\pm 1\}$ composed with the obvious map $\{\pm 1\} \to R$.
- Let G be the quaternion group $Q = \{\pm 1, \pm i, \pm j, \pm k\} \subset \mathbb{H}$ with the multiplication given by that of $\mathbb{H}$. Then $\mathbb{R}[G] \cong \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{H}$ so $S_{\mathbb{R}}(G)$ has 5 elements.

Proposition A.2.1. *If G is a finite abelian group and k is an algebraically closed field, then every simple $k[G]$-module is of dimension 1 over k.*

pf. Let V be a simple $k[G]$-module. Then any nonzero element $v \in V$ gives a surjective k-linear map $k[G] \to V$ sending $x \in k[G]$ to xv. Then V is a finite dimensional vector space, and so is $\mathrm{End}_{k[G]}(V)$. The finite dimensional k-algebra $\mathrm{End}_{k[G]}(V)$ is also a k-division algebra by the Schur's lemma, $\mathrm{End}_{k[G]}(V) = k$. Moreover, as G is abelian, $k[G]$ is a commutative ring, so the action of any element of $k[G]$ on V is $k[G]$-linear. Thus the action of $k[G]$ on V is given by a map of k-algebras $k[G] \to \mathrm{End}_{k[G]}(V) = k \subset \mathrm{End}_k(V)$, So $\forall v \in V$, the subspace kv is a $k[G]$-submodule of V. As V is irreducible, $\dim V = 1$.

e.g. Let $G = \mathbb{Z}/n\mathbb{Z},\ \ n \geq 1,\ \ k$ be an algebraically closed field.

- char $k \not| n$: fix a primitive nth root ζ_n of 1 in k. The k-algebra $k[G]$ is semisimple, so $k[G] = k \times \cdots \times k$, where n factors k correspond to the n irreducible representations of G given by the maps $\begin{array}{rcl} G = \mathbb{Z}/n\mathbb{Z} & \to & k^\times \\ i & \mapsto & \zeta_n^i \end{array}$.
- char $k|n$: let $p = \mathrm{char}(k)$ and write $n = p^r m$ with $(m,p) = 1$. Then $G = \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$. $(1,0),(0,1) \in \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ have images $a, b \in k^\times$ under the map $G \to k^\times$ corresponding to the representation. Then $a^{p^r} = 1$ so $a = 1$ given $\mathrm{char}(k) = p$. Also b can be any of the m solutions of the equation $x^m = 1$ in k. So we can get m irreducible representations of G, and $k[G]/\mathrm{rad}(k[G]) \cong k^m$.

A.2.2 Characters in $\mathrm{char}(k) = 0$

Definition A.2.2. *A function $f : G \to k$ is called central if $\forall g, h \in G, \quad f(gh) = f(hg)$. We write $\mathcal{C}(G,k)$ for the k-algebra of central functions from G to k.*

Definition A.2.3. *Let (V,ρ) be a representation of G. The character of V is the function*

$$\begin{array}{rcl} \chi_V : G & \to & k \\ g & \mapsto & T(\rho(g)) \end{array}$$

Proposition A.2.2. *Let $(V,\rho_V), \quad (W,\rho_W)$ be representations of G. Then* $\left\{ \begin{array}{r} \chi_V(1) = \dim_k V \\ \chi_V \in \mathcal{C}(G,k) \\ \chi_{V\oplus W} = \chi_V + \chi_W \\ \chi_{V\otimes_k W} = \chi_V \chi_W \end{array} \right.$.

pf. Choose k-bases $\left\{ \begin{array}{cc} (e_1,\ldots,e_n) & V \\ (f_1,\ldots,f_m) & W \end{array} \right.$. Let $g \in G$. In the chosen bases of V, W, write $\left\{ \begin{array}{c} \rho_V(g) \\ \rho_W(g) \end{array} \right.$ as matrices $\left\{ \begin{array}{c} (x_{ij}) \\ (y_{ij}) \end{array} \right.$. Then $(e_i \otimes f_j)$ is a basis for $V \otimes_k$

W, and the matrix $\rho_V(g) \otimes \rho_W(g)$ in this basis is $(x_{ii'}y_{jj'})$. Hence $\chi_{V\otimes_k W}(g) = \sum_{i=1}^n \sum_{j=1}^m x_{ii}y_{jj} = (\sum_i x_{ii})(\sum_j y_{jj}) = \chi_V(g)\chi_W(g)$.

Corollary A.2.1. *The map $V \mapsto \chi_V$ induces a morphism of rings $R_k(G) \to \mathcal{C}(G,k)$.*

For K/k an extension of fields, given every representation of G over k, $\chi_{V\otimes_k K} = \chi_V$, so
$$\begin{array}{ccc} R_k(G) & \longrightarrow & \mathcal{C}(G,k) \\ \downarrow & & \downarrow \\ R_K(G) & \longrightarrow & \mathcal{C}(G,K) \end{array}$$
where the left vertical arrow is given by $[V] \mapsto [V \otimes_k K]$ and the right vertical arrow is the obvious inclusion.

Definition A.2.4. *Let V, W be representations of G. The k-vector space $\hom_k(V,W)$ becomes a representation of G if we make $g \in G$ act by $(g\cdot f)(v) = gf(g^{-1}v)$, $\forall f \in \hom_k(V,W)$ and $\forall v \in V$. In particular, $V^* \equiv \hom_k(V,k)$ is a representation of G and $(g \cdot f)(v) = f(g^{-1}v)$, $\forall g \in G$, $f \in V^*$, $v \in V$.*

Definition A.2.5. *For every representation V of G, we set*

$$V^G = \{v \in V | \forall g \in G, \ \ gv = v\}$$

This is the space of invariants of G in V.

$\to \hom_G(V,W) = \hom_k(V,W)^G$

Proposition A.2.3. *Let V, W be representations of G. Then the map* $\begin{array}{rcl} \phi : V^* \otimes_k W & \to & \hom_k(V,W) \\ v \otimes w & \mapsto & (v \mapsto f(v)w) \end{array}$ *is a G-equivariant isomorphism.*

pf. It is well-defined, because the map $\begin{array}{rcl} V \times W & \to & \hom_k(V,W) \\ (v,w) & \mapsto & (v \mapsto f(v)w) \end{array}$ is k-bilinear. Let $f \in V^*$, $w \in W$, $g \in G$, $v \in V$. Then $\left\{ \begin{array}{r} (\phi(g(f\otimes w)))(v) = (\phi((gf)\otimes(gw))(v) = (gf)(v)gw = f(g^{-1}v)gw \\ (g\phi(f \otimes w))(v) = g(\phi(f\otimes w)(g^{-1}v)) = gf(g^{-1}v)w \end{array} \right.$ so ϕ is G-equivariant. Let (e_i) be a basis of W as a k-vector space. Then we have k-linear isomorphisms $\begin{array}{rcl} u : \oplus_i \hom_k(V,k) & \to & \hom_k(V,W) \\ (f_i) & \mapsto & (v \mapsto \sum f_i(v)e_i) \end{array}$, $\begin{array}{rcl} v : \oplus V^* & \to & V^* \otimes_k W \\ (f_i) & \mapsto & \sum f_i \otimes e_i \end{array}$. Then $\phi = u \circ v^{-1}$ is an isomorphism.

Proposition A.2.4. *Let V, W be representations of G. Then $\forall g \in G$,*
$\left\{ \begin{array}{r} \chi_{V^*}(g) = \chi_V(g^{-1}) \\ \chi_{\hom_k(V,W)}(g) = \chi_V(g^{-1})\chi_W(g) \end{array} \right.$.

pf. Write ρ for the action morphism $G \to \mathrm{End}_k(V)$. Choose a basis $\mathcal{B}$ of V as a k-vector space, and let M be the matrix of $\rho(g)^{-1}$ in $\mathcal{B}$.Then g acts by the matrix M' on V^*, so the result follows. Then the first stmt implies the second stmt.

Theorem A.2.3 (Schur's Lemma)**.** *Let V, W be irreducible representations of G. Then $\hom_G(V, W) = 0$ unless $V \cong W$, and $End_G(V)$ is a finite-dimensional k-division algebra. If moreover k is algebraically closed, then $End_G(V) = k$.*

pf. The first part follows from the first Schur's lemma.
We have a map of rings $\begin{array}{ccc} K & \to & \mathrm{End}_G(V) \\ \lambda & \mapsto & \lambda \cdot 1 \end{array}$ which is injective, given k a field. Let $a \in R$. Then $\begin{array}{ccc} m_a : \mathrm{End}_G(V) & \to & \mathrm{End}_G(V) \\ x & \mapsto & ax \end{array}$ is k-linear. Here $\mathrm{End}_G(V)$ is finite dimensional k-vector space and k is algebraically closed, so m_a has at least one eigenvalue, i.e. $\exists \lambda \in k$ s.t. $\ker(m_a - \lambda \cdot Id) = \ker(m_{a-\lambda}) \neq 0$. This implies $a - \lambda$ is not invertible. Since $\mathrm{End}_G(V)$ is a division algebra, $a - \lambda = 0 \Leftrightarrow a = \lambda \in k$.

Lemma A.2.1. *Suppose k is algebraically closed and let (V, ρ) be an irreducible representation of G. Then*

$$\sum_{g \in G} \chi_V(g) = \begin{cases} 0 & V \ncong \nVdash \\ |G| & V \cong \nVdash \end{cases}$$

pf. Extend $\rho : G \to \mathrm{End}_k(V)$ to $\rho : k[G] \to \mathrm{End}_k(V)$ and let $c = \sum_{g \in G} g \in k[G]$. Then c is central in $k[G]$, so $\rho(c) \in \mathrm{End}_k(V)$ is a G-equivariant endomorphism of V. Then by the Schur's lemma, $\exists \lambda \in k$ s.t. $\rho(c) = \lambda id_V$, so $\sum_{g \in G} \chi_V(g) = T(\rho(c)) = \lambda \dim V$. Moreover $\forall h \in G,\ \ \lambda \rho(h) = \rho(c)\rho(h) = \rho(ch) = \rho(c) = \lambda id_V$. So if $\lambda \neq 0,\ \ V \cong \nVdash,\ \ \sum \chi_V(g) = \sum 1 = |G|$.

Lemma A.2.2. *Let V be a representation of G. Then*

$$\sum_{g \in G} \chi_V(g) = |G| \dim_k(V^G)$$

pf. Assume wlog k is algebraically closed (we has seen $\chi_{V \otimes_k K}) = \chi_V$ and we can make V algebraically closed by tensoring with algebraically closed K). Write $V = \oplus_{W \in S_k(G)} W^{\oplus n_W}$. Then $\sum_{g \in G} \chi_V(g) = \sum_{W \in S_k(G)} n_W \sum_{g \in G} \chi_W(g) = n_{\nVdash}|G|$ by the previous lemma.
If $W \in S_k(G),\ \ W \ncong \nVdash,\ \ W^G = 0$ since W^G is a subrepresentation of W. Then $V^G = \nVdash^{n_{\nVdash}},\ \ \dim_k(V^G) = n_{\nVdash}$.

Theorem A.2.4. *Let V, W be representations of G. Then*

$$\frac{1}{|G|} \sum_{g \in G} \chi_V(g^{-1}) \chi_W(g) = \frac{1}{|G|} \sum_{g \in G} \chi_{V^*}(g) \chi_W(g) = \dim_k(\hom_G(V, W))$$

pf. $\frac{1}{|G|}\sum \chi_V(g^{-1})\chi_W(g) = \frac{1}{|G|}\sum \chi_{V^*}(g)\chi_W(g) = \frac{1}{|G|}\sum \chi_{\hom_k(V,W)}(g) = \dim_k(\hom_k(V,W)^G) = \dim_k(\hom_G(V,W))$ by the previous lemma.

Corollary A.2.2. *If V, W are irreducible and nonisomorphic, then $\sum_{g\in G}\chi_{V^*}(g)\chi_W(g) = 0$.*

pf. by the previous theorem and the Schur's lemma.

Corollary A.2.3. *Suppose k is algebraically closed and let V be a representation of G. Then V is irreducible iff $\sum_{g\in G}\chi_{V^*}(g)\chi_V(g) = |G|$.*

pf. By the previous theorem $\sum_{g\in G}\chi_{V^*}(g)\chi_V(g) = |G|\dim_k(\mathrm{End}_G(V))$. If V is irreducible, $\mathrm{End}_G(V) = k$ by Schur's lemma so the one direction follows. Conversely, suppose $\sum_{g\in G}\chi_{V^*}(g)\chi_V(g) = |G|$. Write $v = \oplus_{i\in I}V_i^{\oplus n_i}$ where V_i are irreducible and pairwise nonisomorphic. Then by the previous corollary, $\sum \chi_{V^*}(g)\chi_V(g) = \sum n_i n_j \sum \chi_{V_i^*}(g)\chi_{V_j}(g) = |G|\sum n_i^2$. So only one of the n_i can be nonzero, and moreover it has to be equal to 1, hence V is irreducible.

Corollary A.2.4. *The family $(\chi_V)_{V\in S_k(G)}$ is linearly independent in $\mathcal{C}(G,k)$.*

pf. Suppose $\sum_{W\in S_k(G)}\alpha_W\chi_W = 0$, with $\alpha_W \in k$. Then $\forall V \in S_k(G)$, $0 = \sum_{g\in G}(\sum_{W\in S_k(G)}\alpha_W\chi_W(g))\chi_{V^*}(g)$
$= \sum \alpha_W \sum \chi_W(g)\chi_{V^*}(g) = \alpha_V|G| \to \alpha_V = 0$, so LI follows.

Corollary A.2.5. *For any representations V, V' of G, $\quad V \cong V'$ iff $\chi_V = \chi_{V'}$.*

pf. Write $\begin{cases} V = \bigoplus_{W\in S_k(G)} W^{\oplus n_W} \\ V' = \bigoplus_{W\in S_k(G)} W^{\oplus n'_W} \end{cases}$. Then $\begin{cases} \chi_V = \sum n_W\chi_W \\ \chi_{V'} = \sum n'_W\chi_W \end{cases}$. By the previous corollary, $\chi_V = \chi_{V'}$ iff $n_W = n'_W$, $\forall W \in S_k(G) \Leftrightarrow V \cong V'$.

Lemma A.2.3. *If $f \in \mathcal{C}(G,k)$ is s.t. $\sum_{g\in G} f(g)\chi_{W^*}(g) = 0$, $\forall W \in S_k(G)$, then $f = 0$.*

pf. If $\rho : G \to \mathrm{End}_k(V)$ is a representation of G, we set $\rho(f) = \sum f(g)\rho(g) \in \mathrm{End}_k(V)$. Then $\forall g \in G$, $\rho(g)\rho(f) = \sum_{h\in G} f(h)\rho(gh) = \sum_{h\in G} f(h)\rho(ghg^{-1}g) = \sum f(ghg^{-1})\rho(ghg^{-1})\rho(g) = \rho(f)\rho(g)$ because f is central. So if V is irreducible, $\rho(f) \in \mathrm{End}_G(V) = k$ by Schur's lemma, hence $\rho(f) = \lambda id_V$, $\lambda \in k$. So $\lambda\dim(V) = T(\rho(f)) = \sum_{g\in G} f(g)\chi_V(g) = 0 \to \lambda = 0$ and $\rho(f) = 0$. By the semisimplicity of $k[G]$, $\rho(f) = 0$ for any representation ρ of G. Then $0 = \rho_{reg}(f)1 = \sum f(g)g$ in $k[G]$, so $f(g) = 0$, $\forall g \in G$.

Theorem A.2.5. *Suppose that the field k is algebraically closed. Then the family $(\chi_W)_{W\in S_k(G)}$ is a basis of $\mathcal{C}(G,k)$.*

pf. They are L.I by the corollary A.2.4. so we just need to show it spans $\mathcal{C}(G,k)$. Let $f \in \mathcal{C}(G,k)$. Then $\forall W \in S_k(G)$, $\alpha_W = \frac{1}{|G|}\sum_{g\in G} f(g)\chi_{W^*}(g)$. Set $f' = f - \sum_{W\in S_k(G)} \alpha_W \chi_W$. Then $\forall W \in S_k(G)$, $\sum_{g\in G} f'(g)\chi_{W^*}(g) = \sum f(g)\chi_{W^*}(g) - \alpha_W \sum \chi_W(g)\chi_{W^*}(g) = 0 \to f' = 0$ and f corresponds to χ_W by the previous corollaries. So it spans.

Corollary A.2.6. *Suppose that k is algebraically closed. Then* $|S_k(G)| = \dim_k \mathcal{C}(G,k)$ *and this is also equal to the number of conjugacy classes in* G.

Corollary A.2.7. *Suppose k is algebraically closed.Then the map* $\begin{array}{ccc} R_k(G) & \to & \mathcal{C}(G,k) \\ V & \mapsto & \chi_V \end{array}$ *induces an isomorphism of k-algebra* $R_k(G)\otimes_{\mathbb{Z}} k \to \mathcal{C}(G,k)$.

Let G_1, G_2 be two groups. If V_1/V_2 is a representation of G_1/G_2, then $V_1 \otimes_k V_2$ becomes a representation of $G_1\times G_2$ with action $(g_1,g_2)(v_1\otimes v_2) = (g_1v_1)\otimes(g_2v_2)$.

Theorem A.2.6. *Suppose k is algebraically closed, and let* V_1/V_2 *be a representation of* G_1/G_2. *The representation* $V_1\otimes_k V_2$ *of* $G_1 \times G_2$ *is irreducible iff* V_1, V_2 *are both irreducible. Every irreducible representation of* $G_1\times G_2$ *is of the form* $V_1 \otimes_k V_2$.

pf. The character of $V \equiv V_1 \otimes_k V_2$ is the map $(g_1,g_2)\mapsto \chi_{V_1}(g_1)\chi_{V_2}(g_2)$. So $< V, V >_G = \frac{1}{|G_1\times G_2|}\sum_{(g_1,g_2)} \chi_{V_1}(g_1)\chi_{V_2}(g_2) =< V_1, V_1 >_{G_1}< V_2, V_2 >_{G_2}$. All three brackets are nonnegative integers, $< V, V >_G = 1$ iff $< V_1, V_1 >_{G_1} =< V_2, V_2 >_{G_2} = 1$. So if V_1, V_2 are irreducible representation of G_1/G_2, then $V_1\otimes V_2$ is an irreducible representation of G. Thus we can define a map $S_k(G_1)\times S_k(G_2)\to S_k(G_1\times G_2)$. Since the representations $V_1\otimes V_2 \sim V_1'\otimes V_2'$ iff $V_1 \sim V_1'$, $V_2\sim V_2'$, the map is injective. Let $X(G_1)$ be the set of conjugacy classes in G_1, and similarly for $X(G_2), X(G_1\times G_2)$. Then we have a surjective map $u : G_1\times G_2 \to X(G_1\times G_2)$. Then (g_1,g_2), $(g_1',g_2')\in G_1\times G_2$ are conjugate iff $g_i \sim g_i'$ are conjugate in G_i. So u induces a bijection $X(G_1)\times X(G_2)\to X(G_1\times G_2)$. As $|S_k(G)| = |X(G)|$, $\forall G \in \{G_1, G_2, G_1\times G_2\}$, so $|S_k(G_1)\times S_k(G_2)| = |S_k(G_1\times G_2)|$. So source and target have the same cardinality and the map is a bijection and therefore the theorem follows.

Fix a finite group G and we assume k is any field s.t. char$(k) \not| |G|$. If $f_1, f_2 \in \mathcal{C}(G,k)$, write $< f_1, f_2 >_G = \frac{1}{|G|}\sum_{g\in G} f_1(g)f_2(g^{-1}) = \frac{1}{|G|}\sum_{g\in G} f_1(g^{-1})f_2(g)$. Then the previous theorem can be expressed as; for any representations V, W of G, we have $< \chi_V, \chi_W >= \dim_k(\hom_G(V,W))$.

Definition A.2.6. *Let H be a subgroup of G. If* $f \in \mathcal{C}(H,k)$, *define*
$$\begin{array}{cccc} Ind_H^G f : G & \to & k \\ g & \mapsto & \frac{1}{|H|}\sum_{s\in G|_{s^{-1}gs\in H}} f(s^{-1}gs) \end{array}.$$

Theorem A.2.7. $\begin{array}{c} \forall f \in \mathcal{C}(H,k), \quad Ind_H^G f \in \mathcal{C}(G,k) \\ Ind_H^G \chi_V = \chi_{Ind_H^G V} \end{array}$ *for a representation V of H.*

pf. Let (V, ρ_V) be a representation of H and write $(W, \rho_W) = \mathrm{Ind}_H^G(V, \rho_V)$. Let $g_1, \ldots, g_r$ be a system of representatives of G/H. Then $k[G] = \bigoplus_{i=1}^r g_i k[G]$, so $W = k[G] \otimes_{k[H]} V = \bigoplus_{i=1}^r W_i$ with $W_i = g_i k[H] \otimes_{k[H]} V$. Let $g \in G$. Then $\exists \sigma \in \mathcal{G}_r$ and $h_1, \ldots, h_r \in H$ s.t. $gg_i = g_{\sigma(i)} h_i$, $\forall i$. Then $\rho_W(g)(W_i) = W_{\sigma(i)}, \forall i$. Choose a basis $(e_1, \ldots, e_n)$ of V. Then $\forall i$, $(g_i \otimes e_1, \ldots, g_i \otimes e_n)$ is a basis of W_i, and $\rho_W(g)(g_i \otimes e_s) = g_{\sigma(i)} \otimes \rho_V(h_i) e_s$. Thus $T(\rho_W(g)) = \sum_{i|_{\sigma(i)=i}} T(\rho_V(h_i)) = \sum_{i|_{g_i^{-1} g g_i \in H}} f(g_i^{-1} g g_i)$. But $G = \coprod_{i=1}^r g_i H$ and if $s \in g_i H$, $\left\{ \begin{array}{c} s^{-1} g s \in H \text{ iff } g_i^{-1} g g_i \in H \\ f(s^{-1} g s) = f(g_i^{-1} g g_i) \end{array} \right.$. So the second stmt follows. The first stmt follows by the first one.
$\to$ if $f \in \mathcal{C}(G,k)$, $\mathrm{Res}_H^G f$ for $f|_H$. Then $\mathrm{Res}_H^G \chi_V = \chi_{\mathrm{Res}_H^G V}$.

Theorem A.2.8. *If* $\left\{ \begin{array}{c} f_1 \in \mathcal{C}(H,k) \\ f_2 \in \mathcal{C}(G,k) \end{array} \right.$ *, then* $< f_1, Res_H^G f_2 >_H = < Ind_H^G f_1, f_2 >_G$.

pf. $< f_1, \mathrm{Res}_H^G f_2 >_H = \frac{1}{|H|} \sum_{h \in H} f_1(h) f_2(h)$
$< \mathrm{Ind}_H^G f_1, f_2 >_G = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|H|} \sum_{s \in G|_{s^{-1} g s \in H}} f_1(s^{-1} g s) f_2(g) = \frac{1}{|G|} \sum \frac{1}{|H|} \sum f_1(s^{-1} g s) f_2(s^{-1} g s) = \frac{1}{|G|} \frac{1}{|H|} \sum_{h \in H} \sum_{s \in G} f_1(h) f_2(h) = \frac{1}{|H|} \sum_{h \in H} f_1(h) f_2(h)$. They are equivalent.

Let R be a ring, G be a finite group and H, K be subgroups of G. Let (V, ρ) be a representation of H and $g_1, \ldots, g_r$ be a system of representatives of the double classes in $K \setminus G/H$, i.e. $G = \coprod_{i=1}^r K g_i H$. For every $i \in \{1, \ldots, r\}$, let $H_i = g_i H g_i^{-1} \cap K$ and let (V_i, ρ_i) be the representation of H_i on V, given by $\rho_i(h) = \rho(g_i^{-1} h g_i)$.

Theorem A.2.9 (Mackey's Formula)**.** *We have an isomorphism of $R[K]$-modules*

$$Res_K^G Ind_H^G V \cong \bigoplus_{i=1}^r Ind_{H_i}^K V_i$$

pf. $\forall j \in \{1, \ldots, r\}$, let $(x_i)_{i \in I_j}$ be a system of representatives of K/H_j. Then $\{x_i g_j, 1 \leq j \leq r, i \in I_j\}$ is a system of representatives of G/H $(G = \coprod_{j=1}^r K g_j H = \coprod_{j=1}^r \coprod_{i \in I_j} x_i H_j g_j H = \coprod \coprod x_i g_j (g_j^{-1} H_j g_j) H = \coprod \coprod x_i g_j H$). So $W \equiv \mathrm{Ind}_H^G V = \bigoplus_{j=1}^r W_j$, $W_j = \bigoplus_{i \in I_j} x_i g_j k[H] \otimes_{k[H]} V$. Fix j and consider the R-linear maps $\begin{array}{ccc} \phi : \mathrm{Ind}_{H_j}^K V_j & \to & W_j \\ \sum_{i \in I_j} x_i \otimes v_i & \mapsto & \sum (x_i g_j) \otimes v_i \end{array}$ and $\begin{array}{ccc} \psi : W_j & \to & \mathrm{Ind}_{H_j}^K V_j \\ \sum_{i \in I_j} (x_i g_j) \otimes v_i & \mapsto & \sum_{i \in I_j} x_i \otimes v_i \end{array}$. Then ϕ, ψ are inverses to each other. Since ϕ is K-linear $(\forall y \in K, \exists \sigma \in \mathcal{G}_{I_j}, h_i \in H_j, i \in I_j$ s.t. $y x_i = x_{\sigma(i)} h_i$, $\forall i \in I_j$, $\phi(y(\sum x_i \otimes v_i)) = \phi(\sum (y x_i) \otimes v_i) = \phi(\sum (x_{\sigma(i)} h_i) \otimes v_i) = \phi(\sum x_{\sigma(i)} \otimes (\rho_j(h_i) v_i)) = \phi(\sum x_{\sigma(i)} \otimes (\rho(g_j^{-1} h_i g_j) v_i)) =$

$\sum(x_{\sigma(i)}g_j)\otimes(\rho(g_j^{-1}h_ig_j)v_i)=\sum(x_{\sigma(i)}h_ig_j)\otimes v_i=\sum(yx_ig_j)\otimes v_i=y\phi(\sum x_i\otimes v_i)$), the theorem follows.

$$<V,W>_G=<\chi_V,\chi_W>_G=\dim_k \hom_G(V,W)$$

Let H be a subgroup of G and (V,ρ) be a representation of H. For every $g\in G,\quad H_g=gHg^{-1}\cap H$ and define two representations $\begin{cases}\rho_g(h)=\rho(g^{-1}hg)\\ \mathrm{Res}_g(\rho)(h)=\rho(h)\end{cases}$.

Theorem A.2.10. *TFAE*

1. *$W\equiv Ind_H^G V$ is irreducible*
2. *V is irreducible, and $\forall g\in G\setminus H,\quad \hom_{H_g}(\rho^g, Res_g(\rho))=0$*

pf. $\begin{array}{ll} <W,W>_G=<V,\mathrm{Res}_H^G W>_H & \text{by Frobenius reciprocity}\\ =<V,\bigoplus_{g\in H\backslash G/H}\mathrm{Ind}_{H_g}^H(\rho^g)>_H & \text{Mackey's formula}\\ =\sum_{g\in H\backslash G/H}<V,\mathrm{Ind}_{H_g}^H(\rho^g)>_H & =\sum<\mathrm{Ind}_{H_g}^H(\rho^g),V>_H\\ =\sum<\rho^g,\mathrm{Res}_g(\rho)>_H & \text{by Frobenius reciprocity}\end{array}$ where all the summands are ≥ 0. If $g\in H,\quad H_g=H$ and $\rho_G\cong\mathrm{Res}_g(\rho)=\rho$, so $<\rho^g,\mathrm{Res}_g(V)>_H=<V,V>_H$. Then W is irreducible $\Leftrightarrow<W,W>_G=1\Leftrightarrow<V,V>_H=1$
$<\rho^g,\mathrm{Res}_g(\rho)>_H=0,\forall g\in G\setminus H\Leftrightarrow V$ is irreducible and $\hom_{H_g}(\rho^g,\mathrm{Res}_g(\rho)>=0,\ \forall g\in G\setminus H$.

Proposition A.2.5. *Write $R(G)_{\mathbb{Q}}=R(G)\otimes_{\mathbb{Z}}\mathbb{Q}$. Then the $\mathbb{Z}$-linear map $<\cdot,\cdot>_G$: $R(G)\times R(G)\to\mathbb{Z}$ induces a $\mathbb{Q}$-linear isomorphism*

$$\begin{array}{ccc} R(G)_{\mathbb{Q}} & \to & R(G)^*_{\mathbb{Q}}\equiv\hom_{\mathbb{Q}}(R(G)_{\mathbb{Q}},\mathbb{Q})\\ x & \mapsto & (y\mapsto<x,y>_G)\end{array}$$

Let H be a subgroup of G. If we use the isomorphism above to identify $R(G)_{\mathbb{Q}}$ and $R(H)_{\mathbb{Q}}$ with their duals, then the transpose of Ind_G^H is Res_H^G.

pf.

- Denote by $\begin{cases}(e_V)_{V\in S_k(G)} & \text{the basis } ([V])_{V\in S_k(G)} \text{ of } R(G)_{\mathbb{Q}}\\ (e_V^*)_{V\in S_k(G)} & \text{the dual basis}\end{cases}$. Then by the theorem A.2.4, and the corollary A.2.2., the map above sends each e_V to a nonzero multiple of e_V^* If k is algebraically closed, the map sends each e_V to e_V^* itself by the corollary A.2.3., and it actually induces an isomorphism $R(G)\to\hom_{\mathbb{Z}}(R(G),\mathbb{Z})$.
- by the Frobenius reciprocity

Theorem A.2.11 (Artin)**.** *Let X be a set of subgroups of G. Consider the map $Ind_X=\bigoplus_{H\in X}Ind_H^G:\bigoplus_{H\in X}R(H)\to R(G)$. Then TFAE*

1. $G = \cup_{H\in X} \cup_{g\in G} gHg^{-1}$

2. *$Ind_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is surjective, that is* $\forall x \in R(G),\ \exists d \geq 1,\ \exists x_H \in R(H),\ H \in X,$ *st.* $dx = \sum_{H\in X} Ind_H^G x_H$.

pf.

- Let $S = \cup_{H\in X} \cup_{g\in G} gHg^{-1}$.Then if $H \in X,\ x_H \in R(H)_{\mathbb{Q}},\ \mathrm{Ind}_H^G x_H$ is zero on $G \setminus S$. Then by 2, this implies $x = 0$ on $G \setminus S,\ \forall x \in R(G)_{\mathbb{Q}}$. Then by A.2.5., $\forall f \in \mathcal{C}(G,k),\ f|_{G\setminus S} = 0$. Hence $G \setminus S = \emptyset$.
- By the previous proposition, the transpose of $\mathrm{Ind}_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is $\bigoplus_{H\in X} \mathrm{Res}_H^G : R(G)_{\mathbb{Q}} \to \bigoplus_{H\in X} R(H)_{\mathbb{Q}}$. By 1, the sum of the restriction maps $\mathcal{C}(G,k) \to \bigoplus_{H\in X} \mathcal{C}(H,k)$ is injective, and therefore the map above is injective and 2 follows.

Corollary A.2.8. *Take for X the set of cyclic subgroups of G. Then $Ind_X \otimes_{\mathbb{Z}} \mathbb{Q}$ is surjective.*

pf. Because $\forall g \in G$, it is an element of the cyclic subgroup that it generates, by the Artin's theorem, the corollary follows.

Definition A.2.7. *Let p be a prime number. A finite group H is called p-elementary if $H = C \times P$, with C a cyclic group of order prime to p and P a p-group.*

For every prime number p, let $X(p)$ be the set of p-elementary subgroups of G. Let X be the union of all the $X(p)$ for p prime.

Theorem A.2.12. *Let p be a prime number, write $|G| = p^r m$ with $(p,m) = 1$, and let V_p be the image of*

$$Ind_p \equiv \bigoplus_{H\in X(p)} Ind_H^G : \bigoplus_{H\in X(p)} R(H) \to R(G)$$

Then $m \in V_p$. In particular, $R(G)/V_p$ is a finite group of order prime to p.

pf. Let $n = |G|$ and $\mathcal{O} = \mathbb{Z}[1, \zeta_n, \ldots, \zeta_n^{n-1}] \subset k$, where ζ_n is a primitive nth root of 1 in k. Then $\forall x \in R(G),\ x$ takes values in $\mathcal{O}$. Indeed for every representation (V,ρ) of G and $\forall g \in G,\ \rho(g)$ has eigenvalues in $\mathcal{O}$ by the previous proposition, so $\chi_V(g) \in \mathcal{O}$. We have $\mathcal{O} \cap \mathbb{Q} = \mathbb{Z}$ since $\forall \alpha \in \mathcal{O} \cap \mathbb{Q},\ f \in \mathbb{Q}[T]$ be its minimal polynomial over $\mathbb{Q}$, then $f \in \mathbb{Z}[T]$ as α is integral over $\mathbb{Z},\ \deg f = 1$ as $\alpha \in \mathbb{Q}$. Then the map $\mathcal{O}/\mathbb{Z} \to (\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q})/\mathbb{Q}$ is injective, so $\mathcal{O}/\mathbb{Z}$ is torsion free, so it is free as a $\mathbb{Z}$-module, so $\mathcal{O}$ has a $\mathbb{Z}$-basis of the form $(1, \alpha_1, \ldots, \alpha_c)$. The image of $\mathcal{O} \otimes \mathrm{Ind}_p : \bigoplus_{H\in X} \mathcal{O} \otimes_{\mathbb{Z}} R(H) \to \mathcal{O} \otimes_{\mathbb{Z}} R(G)$ is $\mathcal{O} \otimes_{\mathbb{Z}} V_p$ and we have $(\mathcal{O}\otimes_{\mathbb{Z}} V_p)\cap R(G) = V_p$ since $\mathcal{O}\otimes_{\mathbb{Z}} V_p = V_p \oplus \bigoplus_{i=1}^{c}\alpha_i V_p \subset \mathcal{O}\otimes_{\mathbb{Z}} R(G) = R(G)\oplus\bigoplus \alpha_i R(G)$ and if $y = x+\sum \alpha_i x_i \in \mathcal{O}\otimes_{\mathbb{Z}} V_p,\ \ x\in R(G)$ iff $\forall x_i = 0$. Thus the theorem is reduced to the case when $m\in\mathcal{O}\otimes_{\mathbb{Z}} V_p$.

If C is a cyclic group of G of order c, $\exists$ $\begin{array}{rcl}\theta_C : C & \to & \mathbb{Z} \\ x & \mapsto & \begin{cases} c & C=<x> \\ 0 & otherwise\end{cases}\end{array}$. Let $\theta'_C = \mathrm{Ind}_C^G\theta_C$. Let $x\in G$. Then $\theta'_C(x) = \frac{1}{c}\sum_{s\in G|_{sxs^{-1}\in C}}\theta_C(sxs^{-1}) = \sum_{s\in G|_{<sxs^{-1}>=C}} 1$. $\forall s\in G,\ \ sxs^{-1}$ generates exactly one cyclic subgroup of G. Hence $\sum_{C\subset G,\ cyclic}\theta'_C(x) = \sum_{s\in G} 1 = |G|$. Let C be any cyclic subgroup of G.

- **CLAIM :** $\theta_C \in R(C)$
  pf. Let $c\equiv |C|$. If $c=1,\ \ \theta_C\in R(C)$.
  When $c>1,\quad c = \sum_{B\subset C,\ cyclic}\mathrm{Ind}_B^C\theta_B = \theta_C + \sum_{B\subsetneq C,\ cyclic}\mathrm{Ind}_B^C\theta_B$. Since $c\in R(C)$ and $\forall\theta_B\in R(B),\ \ B\subsetneq C$ by the inductive hypothesis, so $\theta_C\in R(C)$.

Let $f\in\mathcal{C}(G,\mathbb{Z})$ s.t. $f(G)\subset n\mathbb{Z}$. Then write $f = nf',\ f'\in\mathcal{C}(G,\mathbb{Z})$. $n = \sum_{C\subset G,\ cyclic}\mathrm{Ind}_C^G\theta_C$, hence $f = \sum_{C\subset G,\ cyclic}\mathrm{Ind}_C^G(\theta_C)f' = \sum\mathrm{Ind}_C^G(\theta_C\mathrm{Res}_C^G f')$. Write $f_C = \theta_C\mathrm{Res}_C^G f'$. Then $f_C\in\mathcal{C}(C,\mathbb{Z})$ and $f_C(C)\subset |C|\mathbb{Z}$. So $\forall\chi\in R(C),\ <f_C,\chi>_C = \frac{1}{|C|}\sum_{x\in C} f_C(x)\chi(x)\in\mathcal{O}$ and therefore $f_C = \sum_{W\in S_k(C)} <f_C,\chi_W>_C \chi_W\in\mathcal{O}\otimes_{\mathbb{Z}} R(C)$. So $f = \sum_{C\subset G,\ cyclic}\alpha_c\mathrm{Ind}_C^G x_C$, where $\begin{cases}\alpha_c\in\mathcal{O} \\ x_C\in R(C)\end{cases}$.

Definition A.2.8. *$x\in G$ is called p-unipotent if its order is a power of p. $x\in G$ is called p-regular if its order is prime to p.*

- **CLAIM :** $\forall x\in G,\ \ \exists(x_r,x_u)$ **where** $x_r,x_u\in G$ **s.t.** $\begin{cases} x_r \textbf{ is } p\textbf{-regular, and } x_u \textbf{ is } p\textbf{-unipotent} \\ x = x_r x_u = x_u x_r\end{cases}$.
  pf. If x_r, x_u satisfies the conditions above, they have to be powers of x, since if $\begin{cases} a=|x_r| \\ b=|x_u|\end{cases}$, $(a,b)=1$ so there exists an integer $N\geq 1$ s.t. $a|N$ and $N\equiv 1 \mod b$ and then $\begin{cases} x^N = x_r^N x_u^N = x_u \\ x^{1-N} = xx_u^{-1} = x_r\end{cases}$. Suppose we have two such pairs $(x_r,x_u),(x'_r,x'_u)$ satisfying the above conditions. Then $\exists N\geq 1$ s.t. $x_u^{p^N} = (x'_u)^{p^N} = 1,\ \ x_r^{p^N} = x_r,\ \ (x'_r)^{p^N} = x'_r$. Then $x^{p^N} = x_r = x'_r$ and therefore $x_u = x'_u$, so the uniqueness follows. Assume G is generated by x, hence G is cyclic. So we may assume $G = \mathbb{Z}/n\mathbb{Z}$ and $x=1$. Write $n = p^r m$ with $p\not| m$. Then by CRT, $G\cong\mathbb{Z}/p^r\mathbb{Z}\times\mathbb{Z}/m\mathbb{Z}$ and we can take $x_r = (0,1)$ and $x_u = (1,0)$ so such x_r,x_u exist.

- **CLAIM : Let** $\chi\in\mathcal{O}\otimes_{\mathbb{Z}} R(G)$ **be s.t.** $\chi(G)\subset\mathbb{Z}$**, and** $x\in G$**. Write** $x = x_r x_u$**. Then** $\chi(x) = \chi(x_r) \mod p$**.**

pf. Assume $G =< x >$. Let $\chi_1, \dots, \chi_n$ be the characters of the irreducible representations of G over k, which are all 1-dimensional. Write $\chi = \sum_{i=1}^n a_i\chi_i$ with $a_i \in \mathcal{O}$. Let $q = p^r$ be the order of x_i. Then $x^q = x_r^q$ so $\forall i,\ \chi_i(x)^q = \chi_i(x_r)^q$. So $\chi(x)^q = (\sum a_i\chi_i(x))^q = \sum a_i^q\chi_i(x)^q = \sum a_i^q\chi_i(x_r)^q = \chi(x_r)^q \mod p\mathcal{O}$. As $\begin{cases} \chi(x), \chi(x_r) \in \mathbb{Z} \\ p\mathcal{O} \cap \mathbb{Z} = p\mathbb{Z} \end{cases}$, $\chi(x)^q = \chi(x_r)^q \mod p$. Also $a^p \equiv a \mod p,\ \forall a \in \mathbb{Z}$ implies $\chi(x) \equiv \chi(x_r) \mod p$.

- **CLAIM : Let $x \in G$ be p-regular. Let $H = C \times P$ be p-elementary with $C =< x >$ and P a Sylow p-subgroup of $Z_G(x)$. Then $\exists \psi \in \mathcal{O} \otimes_{\mathbb{Z}} R(H)$ s.t. $\psi(H) \subset \mathbb{Z}$ and if $\psi' = \mathrm{Ind}_H^G\psi$,**
$\begin{cases} \psi'(x) \neq 0 \mod p \\ \psi'(s) = 0, \ ,\forall s \in G, \textbf{ a } p\textbf{-regular element that is not conjugate to } x \end{cases}$.

  pf. Let $\begin{cases} c = |C| \\ p^r = |P| \end{cases}$. Let $\begin{array}{rcl} \psi_C : C & \to & \mathbb{Z} \\ y & \mapsto & \begin{cases} c & y = x \\ 0 & otherwise \end{cases} \end{array}$. As $\psi_C(C) \subset c\mathbb{Z}$, $\psi_C \in \mathcal{O}\otimes_{\mathbb{Z}}R(C)$ by the previous claim. Let $\begin{array}{rcl} \psi : H = C \times P & \to & \mathbb{Z} \\ (x, y) & \mapsto & \psi_C(x) \end{array}$.
  Then $\psi \in \mathcal{O}\otimes_{\mathbb{Z}}R(H)$. Let $s \in G$ be p-regular. Then $\psi'(x) = \frac{1}{cp^r}\sum_{y\in G|_{ysy^{-1}\in H}} \psi(ysy^{-1})$. Let $y \in G$. If $ysy^{-1} \in H,\ ysy^{-1} \in C$ so $\psi(ysy^{-1}) \neq 0$ iff $ysy^{-1} = x$. Hence $\psi'(s) = 0$ if s is not conjugate to x. Also $\psi'(x) = \frac{1}{cp^r}\sum_{y\in G|_{yxy^{-1}=x}} \psi(x) = \frac{1}{p^r}\sum_{y\in Z_G(x)} 1 = \frac{1}{p^r}|Z_G(x)| \neq 0 \mod p$.

- **$\exists \psi \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ s.t. $\psi(G) \subset \mathbb{Z}$ and $\psi(x) \neq 0 \mod p,\ \forall x \in G$.**

Write $n \equiv |G| = p^r m$ with $(p, m) = 1$. Choose a $\psi \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ as previously. Let $N = \psi(p^r) = |(\mathbb{Z}/p^r\mathbb{Z})^\times|$. Then $\forall l \in \mathbb{Z}$ prime to p, $l^N = 1 \mod p^r$. So $\forall x \in G,\ \psi(x)^N = 1 \mod p^r$, so $m(\psi^N - 1) \in \mathcal{C}(G, \mathbb{Z})$ takes its values in $n\mathbb{Z}$. Then this implies $m(\psi^N - 1) \in \mathcal{O}\otimes_{\mathbb{Z}}V_p$. AS it is an ideal of $\mathcal{O}\otimes_{\mathbb{Z}}R(G)$, $m\psi^N \in \mathcal{O}\otimes_{\mathbb{Z}}V_p$. Then $m = m\psi^N - m(\psi^N - 1) \in \mathcal{O} \otimes_{\mathbb{Z}} V_p$ and the theorem follows.

Corollary A.2.9. *For every representation V of G, there exist monomial representations $V_1, \dots, V_r$ of G and integers $n_1, \dots, n_r \in \mathbb{Z}$ s.t. in $R(G)$, $[V] = \sum_{i=1}^r n_i[V_i]$.*

Theorem A.2.13 (Brauer)**.** *Let*

$$Ind_X = \bigoplus_{H\in X} Ind_H^G : \bigoplus_{H\in X} R(H) \to R(G)$$

Then it is surjective.

pf. $\mathrm{Im}(\mathrm{Ind}_X) = \sum_{p \text{ prime}} V_p$ by the previous theorem, so $R(G)/\mathrm{Im}(\mathrm{Ind}_X)$ is a finite group of order prime to every prime number, i.e. the trivial group.

Definition A.2.9. *A representation V of G is called monomial if there exists a subgroup H of G and a 1-dimensional representation W of H s.t. $V \cong \mathrm{Ind}_H^G W$.*

Lemma A.2.4. *Let P be a p-group and suppose P is not abelian. Denote by $Z(P)$ the center of P. Then there exists an abelian normal subgroup A of P s.t. $Z(P) \subsetneq A$.*

pf. The quotient $P/Z(P)$ is a nontrivial p-group, so its center is nontrivial. Choose $A' \subset Z(P/Z(P))$ cyclic of order p, and let A be its inverse image in P. Clearly $Z(P) \subsetneq A$ and A is normal in P because it is the inverse image of a normal subgroup of $P/Z(P)$. Also A is abelian because it is generated by $Z(P)$ and by a lift of a generator of A'.

Proposition A.2.6. *Let H be a p-elementary group. Then every irreducible representation of H is monomial.*

pf. Write $H = C \times P$ with C cyclic of order prime to p and P a p-group. Then irreducible representations of H are all of the form $V_1 \otimes_k V_2$ by the theorem A.2.6. where V_1, V_2 is an irreducible representation of C, P respectively. Then V_1 is 1-dimensional, so we just need to show V_2 is monomial. Assume $H = P$ is a p-group. If P is abelian, every irreducible representation of P is 1-dimensional so assume P is not abelian. Let (V, ρ) be an irreducible representation of P. If $\ker\rho \neq \{0\}$, by the inductive hypothesis to $P/\ker\rho$, ρ is monomial. So assume ρ is faithful. Then by the previous lemma, there exists a normal abelian subgroup A of P s.t. $Z(P) \subsetneq A$ where $Z(P)$ is the center of P. Let $V = V_1 \oplus \cdots \oplus V_n$ be the isotypic decomposition of $\mathrm{Res}_A^P V$. Because A is abelian, A acts on each V_i through a morphism of groups $\rho_i : A \to k^\times$. We cannot have $V_1 = V$ because otherwise $\rho(A)$ would be contained in the center of $\rho(P)$, so A would be contained in the center of P, a contradiction. Let $g \in P$, $i \in \{1, \ldots, n\}$. Then if $v \in V_i$, $y \in A$, $\rho(y)\rho(g)v = \rho(g)\rho(g^{-1}yg)v = \rho_i(g^{-1}yg)\rho(g)v$, so $\rho(g)$ sends V_i bijectively to the isotypic component of $\mathrm{Res}_A^P V$ corresponding to the map $\begin{array}{ccc} A & \to & k^\times \\ y & \mapsto & \rho_i(g^{-1}yg) \end{array}$, i.e. the action of P on V permutes the V_i, so we get an action of P on the set $\{V_1, \ldots, V_n\}$. As V is irreducible, this action is transitive. Hence all the V_i are isomorphic as k-vector space and so $\dim_k V_1 = \cdots = \dim_k V_n = \frac{1}{n}\dim_k V$. Let $Q = \{g \in P | \rho(g)V_1 = V_1\}$, then Q is a subgroup of P and $|P/Q| = n > 1$. So $Q \neq P$. As Q stabilises V_1, we get a representation of Q on V_1, denoted by V_Q. Define $\begin{array}{ccc} \phi : \mathrm{Ind}_Q^P V_Q & \to & V \\ g \otimes v & \mapsto & \rho(g)v \end{array}$, which is well-defined by the definition of Q. It is P-equivariant and surjective given $V = \sum_{g \in P} \rho(g)V_1$. Moreover, $\dim_k \mathrm{Ind}_Q^P V_Q =$

$|P/Q| \dim_k V_Q = n \dim_k V_1 = \dim_k V$ so ϕ is an isomorphism and $V \cong \mathrm{Ind}_Q^P V_Q$ as representations of P. Also since V is irreducible, so is V_Q. By the inductive hypothesis, V is monomial.

A.2.3 Characters in $\mathbf{char}(k) = p$

Let R be a ring.

Definition A.2.10. *If M is a R-module, we write $rad(M) = rad(R)M$. This is a submodule of M.*

$\rightarrow$ for R left-Artinian, R-module M is semisimple iff $rad(M) = 0$
pf. If $rad(M) = 0$, M is $R/rad(R)$-module, so it is semisimple because $R/rad(R)$ is a semisimple ring.
If M is semisimple R-module, $M = \bigoplus_{i \in I} M_i$ with all the M_i simple. On each M_i, R acts through $R/rad(R)$ so $rad(R)M = 0$.

Definition A.2.11. *A R-module M is called indecomposable if for every direct sum decomposition $M = M' \oplus M''$, we have either $M' = 0$ or $M'' = 0$.*

Definition A.2.12. *A ring S is called local if $S \neq \{0\}$ and S has a unique maximal left ideal.*

Theorem A.2.14. *Let S be a ring. TFAE*

1. *S is local*
2. *S has a unique maximal right ideal*
3. *$rad(S)$ is a maximal left ideal of S*
4. *$rad(S)$ is a maximal right ideal of S*
5. *$S \neq \{0\}$ and for every $x \in S$, either x or $1 - x$ is invertible*
6. *$S/rad(S)$ is a division algebra*
7. *$S \neq \{0\}$ and every $x \in S \setminus rad(S)$ is invertible*

pf.

- $i \leftrightarrow iii$ follows from the definition of $rad(S)$
  $ii \leftrightarrow iv$ follows since $rad(R) = rad(R^{op})$.
  $vii \leftrightarrow v$ follows since $x \in rad(R)$ and $\forall y, z, \in R$, $1 - yxz$ is invertible are equivalent, when $R, R/rad(R)$ have the same simple modules.
  $vii \rightarrow vi$ trivially

- $iii \to vii$

  pf. Let $x \in S \setminus rad(S)$. Then $Sx \not\subset rad(S)$. As $rad(S)$ is a maximal left ideal of S, $Sx = S$ so $\exists y \in S$ s.t. $yx = 1$. If $xy \in rad(S)$, $1 - yxy = 0$ is invertible, a contradiction. Thus $xy \notin rad(S)$. Then $\exists z \in S$ s.t. $zxy = 1$ so y is a left and right invertible, hence invertible and $x = y^{-1}$ is also invertible.

- $iv \to vii$ follows by the previous argument

- $v \to iii$, $v \to iv$.

  pf. Let $x \in S \setminus rad(S)$. Then $\exists y, z \in S$ s.t. $1 - yxz$ is not invertible. Then by v, this implies yxz is invertible, i.e. $x = y^{-1}z^{-1}$ is also invertible. So any left/right ideal of S that strictly contains $rad(S)$ contains an invertible element, hence is equal to S.

- $vi \to iii$

  pf. Let $x \in S \setminus rad(S)$. Then $\exists y \in S$ s.t. $\begin{cases} yx \in 1 + rad(S) \\ xy \in 1 + rad(S) \end{cases}$. Then xy, yx are invertible hence x, y are invertible. So any left ideal strictly containing $rad(S)$ contains an invertible element, which gives iii.

Proposition A.2.7. *Let S be a left Artinian ring. Then $rad(S)^n = 0$ for n big enough. In fact, if $n = lg(S)$, $rad(S)^n = 0$.*

pf. Let $S = I_0 \supset \cdots \supset I_n = 0$ be a Jordon-Hölder series for S. Then I_i are left ideals of S and I_i/I_{i+1} is a simple S-module $\forall i$. As $rad(S)$ annihilates every simple S-module. $rad(S)I_i \subset I_{i+1}$, $\forall i$ and therefore $rad(S)^n = rad(S)^n I_0 \subset I_n = 0$.

Corollary A.2.10. *Let S be a left Artinian ring. Then TFAE*

1. *S is local*
2. *every element of S is nilpotent or invertible*

pf.

- $i \to ii$

  pf. Let $x \in S$ and suppose x is not nilpotent. Then $x \notin rad(S)$ by the previous proposition, so x is invertible by the previous theorem

- $ii \to i$

  pf. If $x \in S$ is nilpotent, $\sum_{n\geq 0} x^n$ is finite, hence defines an element of S, and this element is the inverse of $1 - x$. So ii implies x or $1 - x$ is invertible $\forall x \in S$, and by the previous theorem S is local.

Proposition A.2.8 (Fitting's Lemma)**.** *Let R be a ring and M be a R-module of finite length. Let $f \in End_R(M)$. Then for n big enough,*

$$M = \ker f^n \oplus Imf^n$$

pf. As M has finite length, every non-increasing or non-decreasing sequence of R-submodules of M has to stabilise. Then for the non-decreasing sequence $\ker f^n$ and the non-increasing sequence $\mathrm{Im} f^n$, we find an integer $N \geq 0$ s.t. $\forall n \geq N$, $\begin{cases} \ker f^n = \ker f^{n+1} \\ \mathrm{Im} f^n = \mathrm{Im} f^{n+1} \end{cases}$. Let $n \geq N$, if $x \in M$, $f^n(x) \in \mathrm{Im} f^{2n}$, so there exists $y \in M$ s.t. $f^n(x) = f^{2n}(y)$ and so $x - f^n(x) \in \ker f^n$ and $x = f^n(x) + (x - f^n(x)) \in \mathrm{Im} f^n + \ker f^n$. So $M = \mathrm{Im} f^n + \ker f^n$. If $\exists 0 \neq x \in \ker f^n \cap \mathrm{Im} f^n$, $x = f^n(y)$, $\exists y \in M$. Then $y \in \ker f^{2n} = \ker f^n$ so $x = f^n(y) = 0$. So $M = \ker f^n \oplus \mathrm{Im} f^n$.

Corollary A.2.11. *Let R, M be as in the Fitting's lemma. Then M is indecomposable iff $End_R(M)$ is local.*

pf. Suppose M is indecomposable, and let $f \in \mathrm{End}_R(M)$. Then there exists an integer $n \geq 1$ s.t. $M = \ker f^n \oplus \mathrm{Im} f^n$. As M is indecomposable, either $\begin{cases} \mathrm{Im} f^n = 0 & f \text{ is nilpotent} \\ \ker f^n = 0 & f^n \text{ is invertible} \end{cases}$.
Suppose M is not indecomposable and write $M = M' \oplus M''$ with $M', M'' \neq 0$. Let $\pi : M \to M'$ be the projection with kernel M''. Then π is not invertible because $\ker \pi = M'' \neq 0$ and $1 - \pi$ is not invertible because $\ker(1 - \pi) = M' \neq 0$ so $\mathrm{End}_R M$ is not local.

Theorem A.2.15 (Krull-Schmidt-Remak Theorem)**.** *Let M be a R-module of finite length. Then $M = M_1 \oplus \cdots \oplus M_r$ with the M_i indecomposable. Moreover, the M_i are uniquely determined up to reordering.*

pf.

- existence

  pf. if $lg(M) = 1$, M is simple and the existence follows trivially.
  if $lg(M) \geq 2$, and M is not indecomposable, write $M = M' \oplus M''$ with $M', M'' \neq 0$. Then $lg(M'), lg(M'') < lg(M)$ and therefore by the inductive hypothesis, we get the decompositions for M', M'' and therefore for M.

- uniqueness

pf. Assume the contrary. Then $M = M_1 \oplus \cdots \oplus M_r = M_1' \oplus \cdots \oplus M_s'$ with all the M_i, M_j' are indecomposable. If $r = 1$, $M_1 = M_1'$ with $s = 1$ is obvious. Suppose $r \geq 2$. Then $\forall j \in \{1, \ldots, s\}$, let $u_j \in \mathrm{End}_R(M_1)$ be the composition $M_1 \hookrightarrow M \to M_j' \hookrightarrow M \to M_1$, where all maps are obvious injections or projections. Then $id_{M_1} = \sum_{j=1}^{s} u_j$, so $\exists j$ s.t. $u_j \notin rad(\mathrm{End}_R(M_1))$. Wlog assume $j = 1$. As M_1 is indecomposable, $\mathrm{End}_R(M_1)$ is local by the previous corollary, so u_1 is invertible by the previous theorem. Write $u_1 = vw$ where w is the composition $M_1 \hookrightarrow M \to M_1'$ and v is the composition $M_1' \hookrightarrow M \to M_1$. Then $(u_1^{-1}v)w = id_{M_1}$ so w is injective and $M_1' = w(M_1) \oplus \ker(u_1^{-1}v)$. As M_1' is indecomposable and $w(M_1) \neq 0$, $\ker(u_1^{-1}v) = 0$, so w is an isomorphism, and so is v. Let $x \in M_1 \cap (\bigoplus_{j \geq 2} M_j')$. Then the projection of x on M_1' is 0 so $w(x) = 0$ so $x = 0$ given w being injective. Hence M_1 and $\bigoplus_{j \geq 2} M_j'$ are direct sums. Moreover if $x \in M$, then the surjectivity of w implies $\exists x_1 \in M_1$ s.t. $x - x_1 \in \sum_{j \geq 2} M_j'$ so $M = M_1 \oplus \sum_{j \geq 2} M_j'$. We get $M = M_1 \oplus \sum_{j \geq 2} M_j'$ and $M_1 \cong M_1'$, by the inductive hypothesis, $M/M_1 \cong \bigoplus_{j=2}^{r} M_j \cong \bigoplus_{j=2}^{s} M_j'$.

Proposition A.2.9. *Let P be a projective R-module, M be a R-module and I be an ideal of R. Then the reduction modulo I map*

$$\hom_R(P, M) \to \hom_R(P/IP, M/IM) = \hom_{R/I}(P/IP, M/IM)$$

is surjective.

pf. Denote by $\begin{cases} \pi : P \to P/IP \\ \pi' : M \to M/IM \end{cases}$ the projections. Then $\forall u \in \hom_R(P/IP, M/IM)$,

$$\begin{tikzcd} P \arrow[r, dotted, "u'"] \arrow[d, "\pi"] \arrow[dr, "u\pi"] & M \arrow[d, "\pi'"] \\ P/IP \arrow[r, "u"] & M/IM \end{tikzcd}$$

. As π' is surjective and P is projective, $\exists u' : P \to M$ s.t. $\pi' u' = u\pi$.

Lemma A.2.5. *If M is a nonzero R-module of finite length, $rad(M) \neq M$.*

pf. Let $M' \subsetneq M$ be a maximal proper submodule. Then M/M' is simple so $rad(R)$ acts trivially on M/M' so $rad(M) = rad(R)M \subset M'$.

Corollary A.2.12. *Assume R is left-Artinian. Let P be a projective R-module of finite length. Then P is indecomposable iff $P/rad(P)$ is simple.*

pf. If P is not indecomposable, $P = P_1 \oplus P_2$, with $P_1, P_2 \neq 0$. Then $P/rad(P) = P_1/rad(P_1) \oplus P_2/rad(P_2)$ and each factor is not zero by the previous lemma. So $P/rad(P)$ is not simple.

Assume $M \equiv P/rad(P)$ is not simple. As R is left-Artinian, $R/rad(R)$ is semisimple, so M is a semisimple module and therefore $M = M_1 \oplus M_2$ with $M_1, M_2 \neq 0$. Let $\pi_1 \in \mathrm{End}_R(M_1)$ be the composition $M \to M_1 \hookrightarrow M$ where the maps are the obvious projection and inclusion. Then by the previous proposition, $\exists \pi \in \mathrm{End}_R(P)$ s.t. $\pi \mod rad(P) = \pi_1$. Then neither π nor $1 - \pi$ are invertible so $\mathrm{End}_R(P)$ is not local, so P cannot be indecomposable by the corollary A.2.11.

Write $PI(R)$ for the set of isomorphism classes of finite length projective indecomposable R-modules and $S(R)$ for the set of isomorphism classes of simple R-modules.

Proposition A.2.10. *Suppose R is left-Artinian and left-Noetherian. Then the map $P \mapsto P/rad(P)$ induces a bijection $PI(R) \to S(R)$.*

pf.

- Well-definedness follows by the previous corollary.
- Let M be a simple R-module. Any $x \in M \setminus \{0\}$ gives a surjective map $\begin{array}{ccc} R & \to & M \\ a & \mapsto & ax \end{array}$. As R is a R-module of finite length, by the Krull-Schmidt-Remak Theorem, we can write $R = P_1 \oplus \cdots \oplus P_n$ where P_i are indecomposable module that are automatically projective as direct summands of a free module. Then $R/rad(R) = \bigoplus P_i/rad(P_i)$ surjects to M and every $P_i/rad(P_i)$ is simple by the previous corollary. So M is isomorphic to one of $P_i/rad(P_i)$ by the Schur's lemma. The map above is surjective.
- Let P, P' be two projective indecomposable modules of finite length, and suppose we have a R-module isomorphism $u : P/rad(P) \to P'/rad(P')$. Then by the previous proposition, there exists a R-module map $u' : P \to P'$ s.t. $u = u' \mod rad(R)$. Let $N \subset P'$ be a proper maximal submodule. Then P'/N is simple so $rad(R)(P'/N) = 0$ so $rad(P') = rad(R)P' \subset N$. As $P'/rad(P')$ is simple, $N = rad(P')$ and $rad(P')$ is the unique proper maximal submodule of P'. As $u'(P) \not\subset rad(P')$, $u'(P) = P'$ so u' is surjective. As P' is projective, $P \cong P' \oplus \ker u'$ but since P is indecomposable, $\ker u' = 0$ and u' is an isomorphism.

If M is a simple R-module, its inverse image in $PI(R)$ is a minimal projective R-module s.t. P surjects onto M. This is called a **projective envelope** or **projective cover of** M.

Theorem A.2.16. *Let S be a ring, and I be an ideal of S s.t. every element of I is nilpotent. Let $\bar{e} \in S/I$ be idempotent. Then there exists $e \in S$ idempotent s.t. $\bar{e} = e \mod I$.*

pf. $\bar{f} = 1 - \bar{e}$ is also idempotent and $\bar{e}\bar{f} = \bar{f}\bar{e} = 0$. Let e be any lift of $\bar{e}$ and let $f = 1 - e$. Then $ef = fe \in I$ and $e + f = 1 \mod I$. Then by the assumption on I, $\exists k \geq 1$ s.t. $(ef)^k = e^k f^k = 0$. Here $e^k = \bar{e}^k = \bar{e} \mod I$. Let $\begin{cases} e' = e^k \\ f' = f^k \end{cases}$. Then $e'f' = f'e' = 0$ and $e' + f' = e^k + f^k = \bar{e} + \bar{f} = 1 \mod I$. Let $x = 1 - (e' + f')$. As $x \in I$, $\exists n \geq 1$ s.t. $x^n = 0$. So $u \equiv 1 + x + \cdots + x^{n-1}$ is an inverse of $1 - x = e' + f'$ and it commutes with e', f'. Let $\begin{cases} e'' = ue' \\ f'' = uf' \end{cases}$. Then $e'' \equiv \bar{e} \mod I$ and $\begin{cases} e''f'' = f''e'' = 0 \\ f'' = uf' \end{cases}$. Then $e'' = \bar{e} \mod I$ so $e''f'' = f''e'' = 0$ and $e'' + f'' = u(e' + f') = 1$. So $(e'')^2 = (e'')^2 + e''f'' = e''(e'' + f'') = e''$ and we have the idempotent lift of $\bar{e}$.

Corollary A.2.13. *Let S be a ring and I be an ideal of S. Suppose that the obvious map $S \to \hat{S} \equiv \lim S/I^n$ is an isomorphism. Then any idempotent of S/I lifts to an idempotent of S.*

pf. Let $\bar{e} \in S/I$ be idempotent. Then by the previous theorem and induction on $n \geq 1$, we can construct a sequence of idempotents $e_n \in S/I^n$ s.t. $e_1 = \bar{e}$ and e_{n+1} lifts to e_n, $\forall n$. Then $e \equiv (e_n) \in \hat{S}$ and its preimage in S is an idempotent lifting $\bar{e}$.

Corollary A.2.14. *1. $PK(R)$ is a free $\mathbb{Z}$-module with basis $([P])_{P \in PI(R)}$.*

2. If P, P' are projective R-modules of finite length, then $P \cong P'$ as R-modules iff $[P] = [P']$ in $PK(R)$.

pf. Write $\begin{cases} P = P_1 \oplus \cdots \oplus P_r \\ P' = P'_1 \oplus \cdots \oplus P'_s \end{cases}$ with P_i, P'_j indecomposable by the Krull-Schmidt-Remak theorem. Then P_i, P'_j are projective since it is a direct summand of projective modules, and therefore they are in $PI(R)$. Then by the same theorem, $P \cong P'$ iff $\exists \sigma : \{1, \ldots, r\} \to \{1, \ldots, s\}$ s.t. $P_i \cong P'_{\sigma(i)}$, $\forall i$. Then by i, this is equivalent to $[P] = [P']$.

Let Λ be a commutative ring and G be a finite group. Assume every $\Lambda[G]$-module is of finite type over Λ.

Definition A.2.13. *We denote by $P_\Lambda(G)$ the quotient of the free $\mathbb{Z}$-module on all the $[P]$, for P projective $\Lambda[G]$-module that is of finite type as a Λ-module, by the relations $[P] = [P'] + [P'']$, for every exact sequence $0 \to P' \to P \to P'' \to 0$.*

$\to \Lambda \to \Lambda'$ a morphism of commutative rings, then $P \mapsto P \otimes_\Lambda \Lambda'$ induces a morphism of groups $P_\Lambda(G) \to P_{\Lambda'}(G)$ (if P is a projective $\Lambda[G]$-module, it is a direct summand of some free $\Lambda[G]$-module F and then $P \otimes_\Lambda \Lambda'$ is a direct summand of the free $\Lambda'[G]$-module $F \otimes_\Lambda \Lambda'$ and the morphism of commutative rings give Λ' as Λ-modules).

Proposition A.2.11. *Let P be a $\Lambda[G]$-module. Then TFAE*

1. *P is a projective $\Lambda[G]$-module*
2. *P is projective as a Λ-module and there exists $u \in End_\Lambda(P)$ s.t. $\forall x \in P,\ \sum_{g \in G} gu(g^{-1}x) = x$.*

pf. Write P_0 for P seen as a Λ-module. Let $Q = \Lambda[G] \otimes_\Lambda P_0$. Then there exists a surjective map $\begin{array}{rcl} q : Q & \to & P \\ x \otimes y & \mapsto & xy \end{array}$.

- **CLAIM :** $\begin{array}{rcl} \phi : \mathbf{End}_\Lambda(P_0) & \to & \hom_{\Lambda[G]}(P, Q) \\ u & \mapsto & \sum_{g \in G} g \otimes ug^{-1} \end{array}$ **is an well-defined isomorphism of Λ-modules.**

  pf. $\phi(u)$ is $\Lambda[G]$-linear $\forall u \in \mathrm{End}_\Lambda(P_0)$ and therefore ϕ is well-defined.
  Let $u \in \mathrm{End}_\Lambda(P_0)$. Here $Q = \bigoplus_{g \in G} g \otimes P_0$ as a Λ-module. So if $x \in P$ s.t. $0 = \phi(u)(x) = \sum g \otimes u(g^{-1}x),\ \ u(g^{-1}x) = 0,\ \forall g \in G$ and in particular $u(x) = 0$. Hence $u = 0$ if $\phi(u) = 0$ and ϕ is injective.
  Let $v \in \hom_{\Lambda[G]}(P, Q)$. Then $v(x) = \sum_{g \in G} g \otimes u_g(x),\ \forall x \in P$ with $u_g \in \mathrm{End}_\Lambda(P_0)$. By $\Lambda[G]$-linearity of v, $\forall h \in G,\ \forall x \in P,\ v(h^{-1}x) = \sum g \otimes u_g(h^{-1}x) = \sum (h^{-1}g) \otimes u_g(x)$ so $u_g(x) = u_{h^{-1}g}(h^{-1}x),\ \forall h, g \in G,\ \forall x \in P$. In particular, $u_g(x) = u_1(g^{-1}x)$,
  $\forall g \in G,\ \forall x \in P$ and therefore $v = \phi(u_1)$ and ϕ is surjective.

- $i \to ii$

  pf. If P is projective, it is a direct summand of some free module $\Lambda[G]^{(I)}$. As $\Lambda[G]$ is a free Λ-module, $\Lambda[G]^{(I)}$ is also a free Λ-module, so P_0 is projective. Also as $q : Q \to P$ is surjective and Λ-linear, $\exists \Lambda$-linear map $s : P \to Q$ s.t. $qs = id_P$. Write $s = \phi(u)$ with $u \in \mathrm{End}_\Lambda(P_0)$. Then $id_P = q \sum gug^{-1}$ which gives ii.

- $ii \to i$

  pf. If P_0 is projective Λ-module, Q is projective $\Lambda[G]$-module. Also $s \equiv \phi(u) : P \to Q$ satisfies $qs = id_P$, so P is a direct summand of Q, hence is also a projective $\Lambda[G]$-module.

Theorem A.2.17. *Suppose Λ is a discrete valuation ring with residue field k and maximal ideal* $\mathbf{m}$.

1. *if P is a $\Lambda[G]$-module that is free of finite type over Λ, then P is projective $\Lambda[G]$-module iff $\bar{P} \equiv P \otimes_\Lambda k$ is a projective $k[G]$-module.*
2. *if P, P' are projective $\Lambda[G]$-modules, then $P \cong P'$ as $\Lambda[G]$-module iff $P \otimes_\Lambda k \cong P' \otimes_\Lambda k$ as $k[G]$-modules.*
3. *suppose the DVR Λ is complete. If $\bar{P}$ is a projective $k[G]$-module, then there exists a unique projective $\Lambda[G]$-module P s.t. $\bar{P} \cong P \otimes_\Lambda k$.*

pf.

- If P is projective as a $\Lambda[G]$-module, then $\bar{P}$ is projective as a $k[G]$-module. Suppose $\bar{P}$ is a projective $k[G]$-module. Then by the previous proposition, $\exists \bar{u} \in \mathrm{End}_k(\bar{P})$ s.t. $\sum g\bar{u}g^{-1} = id_{\bar{P}}$. As P is a projective Λ-module, $\exists u \in \mathrm{End}_\Lambda(P)$ lifting $\bar{u}$ by the proposition A.2.16. Then $u' \equiv \sum gug^{-1} \in \mathrm{End}_{\Lambda[G]}(P)$ is equal to $id_{\bar{P}}$ modulo $\mathbf{m}$. So $\det(u') = 1 \mod \mathbf{m}$, so $\det(u') \in \Lambda^\times$ so u' is invertible and $\sum g(u(u')^{-1})g^{-1} = id_P$. Then by the previous proposition, P is projective $\Lambda[G]$-module.

- Let $\bar{u} : P \otimes_\Lambda k \to P' \otimes_\Lambda k$ be an isomorphism of $k[G]$-modules. Then by the previous proposition A.2.26., there exists a $\Lambda[G]$-module map $u : P \to P'$ lifting $\bar{u}$. We know P, P' are projective Λ-modules of finite type, hence they are free Λ-modules of finite type. Their ranks are equal, because they are equal to the dimension over k of $P \otimes_\Lambda k \cong P' \otimes_\Lambda k$. If we choose Λ-bases of P, P', then the matrix A of u is square, and $\det(A) \neq 0 \mod \mathbf{m}$ because $\bar{u}$ is invertible. Hence $\det(A) \in \Lambda^\times$ and u is an isomorphism of Λ-modules.

- The uniqueness follows from *ii*. For existence, write $\bar{F} = \bar{P} \oplus \bar{P}'$ with $\bar{F} = k[G]^{\oplus n}$, and let $F = \Lambda[G]^{\oplus n}$, and $B = \mathrm{End}_{\Lambda[G]}(F)$. Then the map $B \to \mathrm{End}_{k[G]}(\bar{F})$ is surjective by the proposition A.2.26., so it identifies $\mathrm{End}_{k[G]}(\bar{F})$ with $B/\mathbf{m}B$. Let $\bar{e} \in \mathrm{End}_{k[G]}(\bar{F})$ be the projection on $\bar{P}$ with kernel $\bar{P}'$. Then there exists an idempotent $e \in B$ lifting $\bar{e}$. Then $F = \mathrm{Im}(e) \oplus \ker(e)$ and $P \equiv \mathrm{Im}(e)$ is a projective $\Lambda[G]$-module s.t. $P \otimes_\Lambda k = \mathrm{Im}(\bar{e}) = \bar{P}$.

Fix a complete DVR Λ with uniformizer ω, maximal ideal $\mathbf{m} = (\omega)$, residue field k and fraction field K and assume $char(K) = 0$.

$$\begin{tikzcd}
P_k(G) \arrow[rr, "c"] \arrow[dr, "e"'] & & R_k(G) \\
& R_K(G) \arrow[ur, "d"'] &
\end{tikzcd}$$

where d is the decomposition $P_k(G) \xrightarrow{\psi^{-1}} P_\Lambda(G) \xrightarrow{\otimes_\Lambda K} P_K(G) = R_K(G)$ and c

is the obvious map. Let V be a $K[G]$-module. Let $M_1 \subset V$ be a Λ-lattice, i.e. a finite type Λ-submodule of V s.t. $KM_1 = V$. After replacing M_1 by $\sum_{g\in G} gM_1$, assume M_1 is stable by G. Then $\bar{M_1} \equiv M_1 \otimes_\Lambda k$ is a $k[G]$-module.

Theorem A.2.18. *$[\bar{M_1}] \in R_k(G)$ only depends on V.*

pf. Let M_2 be another G-stable Λ-lattice of V.

- $\omega M_1 \subset M_2 \subset M_1$

  let $N = M_1/M_2$, which is a $k[G]$-module because $\omega M_1 \subset M_2$. From $\omega M_2 \subset \omega M_1 \subset M_2 \subset M_1$, there exists an exact sequence of $k[G]$-modules $0 \to N \to \bar{M_2} \equiv M_2 \otimes_\Lambda k = M_2/\omega M_2 \to \bar{M_1} \to N \to 0$. So in $R_k(G)$, $[N] - [\bar{M_2}] + [\bar{M_2}] - [N] = 0 \to [\bar{M_1}] = [\bar{M_2}]$

- general case

  multiplying M_2 by a high enough power of ω, we may assume $M_2 \subset M_1$. There also exists $n \geq 1$ s.t. $\omega^n M_1 \subset M_2$. Then when n=1, by the previous statement the theorem follows. If $n \geq 2$, let $M_3 = \omega^{n-1}M_1 + M_2$. Then $\omega^{n-1}M_1 \subset M_3 \subset M_1$ so $[M_3 \otimes_\Lambda k] = [\bar{M_1}]$ by the inductive hypothesis. Also $\omega M_3 \subset M_2 \subset M_3$ so $[M_3 \otimes_\Lambda k] = [M_2 \otimes_\Lambda k]$ by the previous statement. Then the theorem holds for n.

Theorem A.2.19. *Assume $p = char(k)$ and $p \not| |G|$.*

1. *every $k[G]$-module is semisimple.*
2. *every $\Lambda[G]$-module that is projective as a Λ-module is projective as a $\Lambda[G]$-module.*
3. *the map $d : R_K \to R_k(G)$ is an isomorphism and so are c, e. Also d induces a bijection $S_K(G) \to S_k(G)$.*

pf.

- i is trivial by the chapter 1 Maschke's theorem
- ii follows from i and the previous theorem A.2.17.
- by i, $P_k(G) = R_k(G)$ and c is just the identity morphism. Also $de = id_{R_k(G)}$ and $e([S_k(G)]) \subset S_K(G)$. Let V, V' be two $K[G]$-modules s.t. $d([V]-[V']) = 0$, let $\begin{cases} M \subset V \\ M' \subset V' \end{cases}$ be G-stable Λ-lattices and let $\begin{cases} \bar{M} = M \otimes_\Lambda k \\ \bar{M'} = M' \otimes_\Lambda k \end{cases}$. Then $d([V] - [V']) = [\bar{M}] - [\bar{M'}] = 0 \in R_k(G)$. As $p \not| |G|$, $\bar{M} \cong \bar{M'}$ as $k[G]$-modules, so $M \cong M'$ as $\Lambda[G]$ by the previous theorem and therefore $V \cong V'$ and $[V] - [V'] = 0$.

CLAIM : Let G be a finite group and p a prime number. If k is algebraically closed field of characteristic p and G is a p-group and V is an irreducible representation of G on a finite dimensional k-vector space, V is the trivial representation.
pf.

- Suppose G is abelian. Let V be an irreducible representation of V. Then by Schur's lemma, $\mathrm{End}_{k[G]}(V) = k$. As G is abelian, $k[G]$ is commutative, so the action of any element of $k[G]$ on V is in $\mathrm{End}_{k[G]}(V)$, so there exists a k-algebra map $\begin{array}{rcl} u : k[G] & \to & k \\ x & \mapsto & (v \mapsto u(x)v) \end{array}$. In particular, every k-subspace of V is a subrepresentation; as V is irreducible $\dim V = 1$. Giving u is the same as giving a morphism of groups $\rho : G \to k^\times$. But the only element of $k^\times$ that has order a power of p is 1, so ρ is trivial so G acts trivially on V.

- Assume G is not abelian. Let Z be the center of G which is a nontrivial abelian p-group, so the inductive hypothesis applies to it. Let V be an irreducible representation of G. Denote by V^Z the k-subspace of $v \in V$ s.t. $gv = v,\ \forall g \in Z$. If $g \in G,\ v \in V^Z$, then $\forall h \in Z,\ h(gv) = (hg)v = (gh)v = g(hv) = gv \in V^Z$. Thus V^Z is a subrepresentation of V. Let $V = V_0 \supset \cdots \supset V_n = 0$ be a Jordan-Hölder series for V seen as a $k[Z]$-module. Then V_{n-1} is a simple $k[Z]$-module, so it is the trivial representation of Z by the inductive hypothesis. This means $V_{n-1} \subset V^Z$ so $V^Z \neq 0$. As V is irreducible $V^Z = V$ so the action of G on V factors G/Z and we can see V as an irreducible representation of G/Z. By the inductive hypothesis, this is the trivial representation of G/Z and so V is the trivial representation of G.

Theorem A.2.20. *The maps*

$$\begin{cases} Ind \equiv \bigoplus_{l,\ prime} \bigoplus_{H \in X(l)} Ind_H^G : \bigoplus_{l \neq p,\ prime} \bigoplus_{H \in X(l)} R_k(H) \to R_k(G) \\ Ind \equiv \bigoplus_{l,\ prime} \bigoplus_{H \in X(l)} Ind_H^G : \bigoplus_{l \neq p,\ prime} \bigoplus_{H \in X(l)} P_k(H) \to P_k(G) \end{cases}$$

are both surjective.

pf. Let $\begin{cases} \mathbb{1}_K & \text{be the unit element in } R_K(G) \\ \mathbb{1}_k & \text{be the unit element in } R_k(G) \end{cases}$. Then $d(\mathbb{1}_K) = \mathbb{1}_k$. By the Brauer's theorem, $\mathbb{1}_K = \sum_{l,\ prime} \sum_{H \in X(l)} \mathrm{Ind}_H^G x_H,\ \exists x_H \in R_K(H)$. Applying d to this equality gives $\mathbb{1}_k = \sum_{l,\ prime} \sum_{H \in X(l)} \mathrm{Ind}_H^G x'_H$ with $x'_H = d(x_H)$ in $R_k(H)$. So if $y \in \begin{cases} R_k(G) \\ P_k(G) \end{cases}$, $y = y\mathbb{1}_k = \sum_{l,\ prime} \sum_{H \in X(l)} \mathrm{Ind}_H^G (x'_H \mathrm{Res}_H^G y)$ where $x'_H \mathrm{Res}_H^G(y) \in P_k(H)$ if $y \in P_k(G)$ given Res_H^G sending $P_k(G)$ to $P_k(H)$.
$\to$ given k algebraically closed, when $l = p$, for any l-elementary group $G = H \times P$,

and P acts simple $k[G]$-module, so each is pulled back from H. Since H is cyclic of order prime to p, it is l'-elementary for any prime $l' \mid |H|$; hence every class $\mathrm{Ind}_H^G[M]$ already lies in the image without $l \neq p$, so domain can be taken as above.

Lemma A.2.6. *Suppose $G = P \times H$ with P a p-group and H of order prime to p. Then P acts trivially on every semisimple $k[G]$-module.*

pf. Let M be a simple $k[G]$-module. As P is a p-group its only irreducible representation over k is the trivial representation, so $M^P \neq 0$. As $G = P \times H$, the action of G preserves M^p. As M is simple, $M^P = M$ and so P acts trivially on M.

Corollary A.2.15. *If k is algebraically closed and K contains all the $|G|$th roots of the unity in $\bar{K}$, then $d : R_K(G) \to R_k(G)$ is surjective.*

pf. By the previous theorem, we may assume G is l-elementary, for some prime l. Write $G = C \times G'$, with C cyclic of order prime to l and G' a l-group. Let $\begin{cases} l = p, & P = G',\ H = C \\ l \neq p & P = C_p,\quad H = C^p \times G' \end{cases}$ where $C = C_p \times C^p$, C_p a p-group, and C^p of order prime to p. Then $G = H \times P$ with P a p-group and H of order prime to p. Let M be a simple $k[G]$-module. Then by the lemma, P acts trivially on M. So M is a simple $k[H]$-module. Then by the previous theorem A.2.17., there exists $K[H]$-module V s.t. $d[V] = [M]$. We see V as a $K[G]$-module by making P act trivially, and then we still have $d[V] = [M]$.

Proposition A.2.12. *If G is a finite p-group, then $\begin{cases} P_k(G) \cong \mathbb{Z} \\ R_k(G) \cong \mathbb{Z} \end{cases}$ and the map $P_k(G) \to R_k(G)$ corresponds to multiplication by $p^n \equiv |G|$.*

pf. The $\mathbb{Z}$-rank of $P_k(G)$ and $R_k(G)$ is 1, and there exists an isomorphism $\begin{array}{ccc} R_k(G) & \to & \mathbb{Z} \\ M & \mapsto & \dim_k M \end{array}$ and $k[G]/rad(k[G]) = k$. Thus $k[G]$ is local and every element of $k[G]$ is nilpotent or invertible. In particular, the only idempotents are 0 and 1. Let $k[G] \equiv M = M_1 \oplus M_2$ with $\begin{cases} M_1 = MI_1 \\ M_2 = MI_2 \end{cases}$ two left ideals $I_1, I_2 \subset k[G]$. Then we get two idempotents $e_1, e_2 \in k[G]$ s.t. $I_1 = k[G]e_1, I_2 = k[G]e_2$, but there are only 0,1 for idempotents, this implies $k[G]$ is a projective indecomposable module, i.e. there is the unique projective indecomposable finite length $k[G]$-module that surjects to $\Bbbk$. Since in $R_k(G)$, $[k[G]] = \dim_k(k[G])\Bbbk = p^n\Bbbk$, the proposition follows.

Theorem A.2.21. *Assume k is algebraically closed. Then $c : P_k(G) \to R_k(G)$ is injective and its image contains $p^n R_k(G)$ where p^n is the biggest power of p dividing $|G|$.*

pf. By the Brauer's theorem we wlog assume G is l-elementary for some prime number l. Then we can write $G = H \times P$, with P a p-group, H of order prime to p. Then the trivial $k[H]$-module is projective given $k[H]$ being a semisimple ring. Thus with trivial action of H, $k[P]$ is a projective $k[G]$-module and by the previous proposition $k[P] = |P|\mathbb{1} \in R_k(G)$ is in the image of c. As $\mathrm{Im}(c)$ is an ideal of $R_k(G)$, , $p^n R_k(G) \subset \mathrm{Im}(c)$.
Since $R_k(G)/\mathrm{Im}(c)$ is a torsion group and $P_k(G), R_k(G)$ are free $\mathbb{Z}$-modules of the same finite rank, c needs to be injective.

Corollary A.2.16. *The map $e : P_k(G) \to R_K(G)$ is injective.*

pf. by the injectivity of c from the previous theorem and the surjectivity of d and commutative diagram above.

Definition A.2.14. *An element $x \in G$ is called p-singular if x is not p-regular, i.e. if p divides the order of x.*

A.3 Compact Group Representation

Definition A.3.1. *A topological group is a topological space G with a group structure s.t. the maps* $\begin{array}{ccc} G & \to & G \\ x & \mapsto & x^{-1} \end{array}$, $\begin{array}{ccc} G \times G & \to & G \\ (x, y) & \mapsto & xy \end{array}$ *are both continuous.*

Let G be a Hausdorff locally compact topological group. By a measure on G, we mean a measure on the Borel σ-algebra of B.

Definition A.3.2. *A nonzero measure μ on G is called a left Haar measure if for every measurable function $f : G \to \mathbb{C}$, $\forall g \in G$,*

$$\int_G f(x) d\mu = \int_G f(gx) d\mu$$

Denote by $\mathcal{C}$ the space of continuous functions with compact support $G \to \mathbb{C}$,equipped with the compact open topology (generated by the sets $\{f \in \mathcal{C} | f(K) \subset U\}$ for $\left\{ \begin{array}{ll} K \subset G & \text{compact} \\ U \subset \mathbb{C} & \text{open} \end{array} \right.$). Let $\mathcal{C}^*$ be the space of continuous linear maps $\mathcal{C} \to \mathbb{C}$, equipped with the weak topology (generated by $\{\lambda \in \mathcal{C}^* : |(\lambda - \lambda_1)(f_1)| < r_1, \dots, |(\lambda - \lambda_n)(f_n)| < r_n\}$, $\forall \lambda_i \in \mathcal{C}^*$, $\forall f_i \in \mathcal{C}$, $\forall r_n \in \mathbb{R}_{++}$). If G is compact and metrisable, for every probability measure μ on G, the linear form $f \mapsto \int_G f(x) d\mu$ on $\mathcal{C}$ is continuous, hence in $\mathcal{C}^*$. This implies we can the set $\mathcal{P}(G)$ of probability measures on G with a subset of $\mathcal{C}^*$.

- **The compact open topology on $\mathcal{C}$ is the same as the topology of uniform convergence, i.e. the topology induced by the norm $||\cdot||_\infty$.**

  pf. Let $f_0 \in \mathcal{C}$
  Assume the open nhood of f_0 is of the form $X \equiv \{f \in \mathcal{C} : f(K_1) \subset U_1, \dots, f(K_m) \subset U_m\}$ where $\forall K_i \subset G$ are compact and $\forall U_j \subset \mathbb{C}$ are open. Then $\forall \epsilon > 0,\ V_\epsilon \equiv \{f \in \mathcal{C} : ||f - f_0||_\infty < \epsilon\}$. Let $i \in \{1, \dots, m\}$. As $f_0(K_i) \subset U_i$ is compact, $\exists \epsilon_i > 0$ s.t. $\{x \in \mathbb{C} : \exists y \in f_0(K_i),\ \ |x-y| < \epsilon\} \subset U_i$. Let $\epsilon = \min_i \epsilon_i$. Then $V_\epsilon \subset X$.
  Let $\epsilon > 0$ and V_ϵ be defined as above. Let $\eta = \epsilon/2$. Then $\forall x \in G,\ U_x \equiv \{y \in G : |f_0(x) - f_0(y)| < \eta\} \subset K_x \equiv \{y \in G : |f_0(x) - f_0(y)| \leq \eta\}$. Then U_x is open and K_x is compact. As $G = \cup_{c \in g} U_x$ is compact, $\exists x_1, \dots, x_n \in G$ s.t. $G = \cup_{i=1}^n U_{x_i} = \cup_{i=1}^n K_{x_i}$. $\forall i,\ \ K_i = K_{x_i}$ for simplicity and let $U_i = \{a \in \mathbb{C} : |f_0(x_i) - a| < \eta\}$. Then $X \equiv \{f \in \mathbb{C} : \forall i,\ \ f(K_i) \subset U_i\} \subset V_\epsilon$.

- **Let ρ be the representation of G on $\mathcal{C}$ defined by $(\rho(g)f)(x) = f(g^{-1}x),\ \forall f \in \mathcal{C},\ \ x, g \in G$. Then $\begin{array}{ccc} G \times \mathcal{C} & \to & \mathcal{C} \\ (g, f) & \mapsto & \rho(g)f \end{array}$ is continuous.**

  pf. Denote by μ the map $G \times \mathcal{C} \to \mathcal{C}$. Let $\left\{ \begin{array}{c} g_0 \in G \\ f_0 \in \mathcal{C} \end{array} \right.$. As G is compact, $f_0 : G \to \mathbb{C}$ is uniformly continuous, so there exist two collections $\left\{ \begin{array}{c} U_1, \dots, U_n \\ V_1, \dots, V_n \end{array} \right.$ of open subsets of G s.t. $\left\{ \begin{array}{r} K_i \equiv \overline{U}_i \subset V_i,\ \ \forall i \\ G = \cup_{i=1}^n U_i \text{ and } \forall i,\ \ \forall x, y \in V_i,\ \ |f_0(x) - f_0(y)| < \epsilon/2 \end{array} \right.$.
  Let $i \in \{1, \dots, n\}$. $\forall x \in K_i,\ \ \exists j$ s.t. $g_0^{-1}x \in V_j$ and open nhood $\left\{ \begin{array}{r} U_x \text{ of } x \text{ in } G \\ W_x \text{ of } g_0 \text{ in } G \end{array} \right.$ s.t. $\forall g \in W_x,\ \forall y \in U_x,\ \ g^{-1}y \in V_j$. As K_i is compact, we can choose $x_1, \dots, x_m$ s.t. $K_i \subset U_{x_1} \cup \dots \cup U_{x_m}$. Then the open nhood $W_i \equiv W_{x_1} \cap \dots \cap W_{x_m}$ of g_0 has the property that $\forall g \in W_i,\ \forall x \in K_i,\ \exists j \in \{1, \dots, n\}$ s.t. $g_0^{-1}x, g^{-1}x \in V_j$. Define $W \equiv \cap_{i=1}^n W_i$. Let $g \in W$ and $f \in \mathcal{C}$ s.t. $||f - f_0||_\infty < \epsilon/2$. Then $\forall x \in G$, choose i s.t. $x \in K_i$. Then $\exists j \in \{1, \dots, n\}$ s.t. $g_0^{-1}x, g^{-1}x \in V_j$. Then $|(gf - g_0f_0)(x)| = |f(g^{-1}x) - f_0(g_0^{-1}x)| = |(f(g^{-1}x) - f_0(g^{-1}x)) + (f_0(g^{-1}x) - f_0(g_0^{-1}x))| \leq |(f(g^{-1}x) - f_0(g^{-1}x))| + |((f_0(g^{-1}x) - f_0(g_0^{-1}x))| < \epsilon/2 + \epsilon/2 = \epsilon$, so it is continuous.

- **The contragredient representation ρ^* s.t. $\rho^*(g)\lambda = \lambda \circ \rho(g^{-1}),\ \forall g \in G,\ \ \lambda \in \mathcal{C}^*$, is continuous.**

  pf. Let $g_0 \in G,\ \ \lambda_0 \in \mathcal{C}^*,\ \ f \in \mathcal{C},\ \ r \in \mathbb{R}_{++}$. Let $||\lambda_0|| = \sup\{||\lambda_0(f_1)||_\infty,\ \ f_1 \in \mathcal{C},\ \ ||f_1||_\infty = 1\}$. Then $||\lambda_0|| < \infty$ since λ_0 is continuous and $||\lambda_0(f_1)||_\infty \leq ||\lambda_0|| ||f_1||_\infty,\ \ \forall f_1 \in \mathcal{C}$. Then by the previous statement, there exists a nhood

W of g_0 in G s.t. $\forall g \in W,\ \ ||g^{-1}f - g_0^{-1}f||_\infty < \frac{r}{2(1+||\lambda_0||)}$. Let $U \equiv \{\lambda \in \mathcal{C}^* : |(\lambda - \lambda_0)(g^{-1}f)| < r/2\}$. Then U is a nhood of λ_0 in $\mathcal{C}^*$ and $\forall g \in W,\ \ \forall \lambda \in U,\ \ |(g\lambda - g_0\lambda_0)(f)| = |\lambda(g^{-1}f) - \lambda_0(g^{-1}f)| + |\lambda_0(g^{-1}f) - \lambda_0(g_0^{-1}f)| < r$.

- **$\mathcal{P}(G)$ is a convex compact G-invariant subset of $\mathcal{C}^*$.**

  pf. Denote by B its unit ball of $\mathcal{C}$ for the sup-norm. Let B_+ be the set of $f \in B$ s.t. $f(G) \subset \mathbb{R}_+$. Then B_+ generates $\mathcal{C}$ as a vector space so the map $\begin{array}{ccc} \mathcal{C}^* & \to & \Pi_{B_+}\mathbb{C} \\ \lambda & \mapsto & (\lambda(f))_{f\in\mathcal{C}} \end{array}$ is injective, and we can use it to identify $\mathcal{C}^*$ to a subspace of $\Pi_{B_+}\mathbb{C}$. By the definition, the weak topology on $\mathcal{C}^*$ is the topology induced by the product topology on $\Pi_B\mathbb{C}$. Let $K = \Pi_B[0,1] \subset \Pi_B\mathbb{C}$. Then by Tychonoff's theorem, K is compact. By the Riesz representation theorem, $\mathcal{C}^* \cap K$ is closed in K. Indeed, if $(a_f)_{f\in B_+}$ is in the closure of $\mathcal{C}^* \cap K$, $\begin{array}{ccc} B_+ & \to & \mathbb{C} \\ f & \mapsto & a_f \end{array}$ extends to a linear map $\mathcal{C} \to \mathbb{C}$ and this linear map is positive, given $a_f \in \mathbb{R}_+$. Thus it is of the form $f \mapsto \int f d\mu$, where μ is a regular Borel measure on G and hence it is continuous, i.e. an element of $\mathcal{C}^*$. So $K \cap \mathcal{C}^*$ is compact and therefore $\mathcal{P}(G)$ is compact given $\mathcal{P}(G) = \{\lambda \in K \cap \mathcal{C}^* | \lambda(1) = 1\}$ is closed in $L \cap \mathcal{C}^*$. The other properties are trivial.

- **G has a left Haar measure.**

  pf. $\mathcal{P}(G)$ is nonempty because $f \mapsto f(1)$ is in $\mathcal{P}(G)$. $\mathcal{C}^*$ is locally convex since $\{\lambda \in \mathcal{C}^* : |\lambda(f_1) < r_1, \ldots, |\lambda(f_s)| < r_s\},\ \ \forall f_i \in \mathcal{C},\ \ \forall r_i \in \mathbb{R}_{++}$ form a basis of nhoods of 0 in $\mathcal{C}^*$ and they are convex. Also the action of G on $\mathcal{P}(G)$ is equicontinuous, i.e. for every nhood of W of 0 in $\mathcal{C}^*$, there exists a nhood V of 0 in $\mathcal{C}^*$ s.t. $\forall \mu_1, \mu_2 \in \mathcal{P}(G)$ s.t. $\mu_1 - \mu_2 \in V,\ \ g \cdot (\mu_1 - \mu_2) \in W,\ \forall g \in G$. Then by the Markov-Kakutani fixed point theorem, $\exists \mu \in \mathcal{P}(G)$ s.t. $g\mu = \mu,\ \forall g \in G$. This is a left Haar measure.

Theorem A.3.1. *Let G be a compact Hausdorff topological group. Let $\mathcal{C}(G,\mathbb{C})$ be the $\mathbb{C}$-algebra of continuous functions from G to $\mathbb{C}$ with the norm $||\cdot||_\infty$, the supremum norm. Then there exists a unique $\mathbb{C}$-linear map $\lambda : \mathcal{C}(G,\mathbb{C}) \to \mathbb{C}$ s.t.* $\left\{ \begin{array}{r} \lambda \textit{ is positive, i.e. } \lambda(f) \geq 0,\ \forall f(G) \subset \mathbb{R}_+ \\ \lambda \textit{ is left invariant, i.e. } \lambda(f) = \lambda(f(g\cdot)),\ \ \forall f \in \mathcal{C}(G,\mathbb{C}), \forall g \in G \\ \lambda \textit{ is right invariant, i.e. } \lambda(f) = \lambda(f(.g)),\ \ \forall f \in \mathcal{C}(G,\mathbb{C}), \forall g \in G \\ \lambda(1) = 1 \end{array} \right.$ *.Moreover, if λ is continuous, there exists a unique probability measure dg on G s.t. $\forall f \in$*

$\mathcal{C}(G,\mathbb{C}),\ \ \lambda(f)=\int_G f(g)dg$ *and this measure satisfies*

$$\int_G f(g)dg = \int_G f(g^{-1})dg \quad \forall f : G \to \mathbb{C}, \textit{measurable.}$$

$\to dg$ is called a bi-invariant probability Haar measure on G.

There exists a function $c : G \to \mathbb{R}_{++}$ **s.t** $\forall g \in G,\ \ \forall f \in \mathcal{C}(G,\mathbb{C}),\ \ \int_G f(xg)d\mu(x) = c(g)\int_G f(x)d\mu(x)$**.**

pf. Let $g \in G$. Then $d\mu(x),\ d\mu(xg^{-1})$ are both left Haar measures on G so $\exists c(g) \in \mathbb{R}_{++}$ s.t. $d\mu(xg^{-1}) = c(g)d\mu(x)$ i.e. $\forall f \in \mathcal{C}(G,\mathbb{C})$, by the previous theorem uniqueness and therefore the above equation holds.
$\to \forall g_1, g_2 \in G,\ \ c(g_1g_2)d\mu(x) = d\mu(x(g_1g_2)^{-1}) = d\mu(xg_2^{-1}g_1^{-1}) = c(g_1)d\mu(xg_2^{-1}) = c(g_1)c(g_2)d\mu(x)$ so $c(g_1g_2) = c(g_1)c(g_2)$, hence $c : G \to (\mathbb{R}_{++},\times)$ is a morphism of groups. Let $\epsilon > 0$. Choose $f \in \mathcal{C}(G,\mathbb{R}_+)$ s.t. $\int_G f(x)d\mu(x) = 1$. Let U be a nhood of 1 in G s.t. $\forall x \in G,\ \ \forall y \in U,\ \ |f(x) - f(xy)| \le \epsilon$ and let K be the suport of f. Then if $g, g' \in G$ s.t. $g^{-1}g' \in U,\ \ |c(g') - c(g)| = |(c(g') - g(g))\int_G f(x)d\mu(x)| = |\int_G f(xg')d\mu(x) - \int_G f(xg)d\mu(x)| \le \epsilon\mu(K)$ hence it is continuous.

Definition A.3.3. *The function* $c : G \to (\mathbb{R}_{++},\times)$ *is called the modular function of* G *and we say* G *is unimodular if* $c(G) = 1$.

If G is compact, it is unimodular.

pf. If G is compact, $c(G)$ is a compact subgroup of $\mathbb{R}_{++}$. The only compact subgroup of $\mathbb{R}_{++}$ is $\{1\}$, so $c(G) = \{1\}$, i.e. G is unimodular.

Let $f \in \mathcal{C}(G,\mathbb{C})$. Then $\int_G f(x^{-1})d\mu(x) = \int_G c(x)f(x)d\mu(x)$.

pf. $\int_G\int_G g(yx)c(x)^{-1}f(x^{-1})d\mu(x)d\mu(y)$

$$= \begin{cases} \int_G f(x^{-1})c(x)^{-1}(\int_G g(yx)d\mu(y))d\mu(x) = \int_G f(x^{-1})d\mu(x)\int_G g(y)d\mu(y) & \text{by the definition of } c \\ \int_G\int_G g(z)c(y^{-1}z)^{-1}f(z^{-1}y)d\mu(z)d\mu(y) = \int_G g(z)d\mu(z)\int_G c(t)f(t)d\mu(t) & \text{by } z = yx,\ \ t = z^{-1}y \end{cases}$$

So the equivalence follows by choosing $\int_G g(x)d\mu(x) = 1$.

If G is just locally compact, we can find nonzero $\mathbb{C}$-linear map satisfying the first and second or the first and third and they are unique upto multiplication by a positive real number. These are also continuous and the corresponding measures on G are called left-invariant or right-invariant Haar measure. Then $\int_G f(g)d_rg = \int_G f(g^{-1})d_lg$, $\begin{cases} d_l & \text{left invariant} \\ d_r & \text{right invariant} \end{cases}$ for every measurable function $f : G \to \mathbb{C}$.

e.g.

- a finite group G with the discrete topology is a topology group. A left and right-invariant Haar measure on G is given by $dg(A) = |A|/|G|$.

- $G = U(1) \equiv \{z \in \mathbb{C} : |z| = 1\}$ with the topology induced by that of $\mathbb{C}$. Then it is a topological group and we have an isomorphism of topological groups $\begin{array}{rcl}\phi : \mathbb{R}/\mathbb{Z} & \to & U(1) \\ t & \mapsto & e^{2i\pi t}\end{array}$. Then $\int_G f(g)dg = \int_0^1 (f \circ \phi)(t)dt = \int_0^1 f(e^{2i\pi t})dt$ where dt is the Lebesgue measure on $\mathbb{R}$.

Definition A.3.4. *Let G be a topological group and V be a normed $\mathbb{C}$-vector space. Then a continuous representation of G on V is an abstract representation of G on V s.t. the action map* $\begin{array}{rcl}G \times V & \to & V \\ (g, v) & \mapsto & gv\end{array}$ *is continuous.*

$\to$ let $\mathrm{End}(V)$ be the $\mathbb{C}$-algebra of continuous endomorphisms of V. Put the operator norm on $\mathrm{End}(V)$ and consider a continuous representation of G on V. Then the action of each $g \in G$ on V is continuous endomorphism of V, so $G \to \mathrm{End}(V)$ exists. But it is not necessarily continuous.

Definition A.3.5. *If V is a normed $\mathbb{C}$-vector space, we denote by $End(V)$ the $\mathbb{C}$-algebra of continuous endomorphisms of V and we put on it the topology given by the operator norm. Write $GL(V)$ for $End(V)^\times$, with the topology induced by that of $End(V)$.*

Proposition A.3.1. *Let V be a normed $\mathbb{C}$-vector space and $\rho : G \to GL(V)$ be a morphism of groups. Consider the following conditions:*

1. *the map* $\begin{array}{rcl}G \times V & \to & V \\ (g, v) & \mapsto & \rho(g)v\end{array}$ *is continuous*

2. *$\forall v \in V$, the map* $\begin{array}{rcl}G & \to & V \\ g & \mapsto & \rho(g)(v)\end{array}$ *is continuous*

3. *the map $\rho : G \to GL(V)$ is continuous*

Then $3 \to 1 \to 2$. If moreover V is finite-dimensional, then all three conditions are equivalent.

pf.

- $1 \to 2$ is trivial.

- $3 \to 1$

  Let $\begin{cases} g_0 \in G \\ v_0 \in V \end{cases}$ and $\epsilon > 0$. Choose a δ s.t. $0 < \delta \leq \frac{\epsilon}{2||\rho(g_0)||}$ and let U be a nhood of g_0 in G s.t. $g \in U$ implies $||\rho(g) - \rho(g_0)|| < \frac{\epsilon}{2(||v_0||+\delta)}$. Then if $\begin{cases} g \in U \\ ||v - v_0|| < \delta \end{cases}$,

  $$\begin{aligned} ||\rho(g)(v) - \rho(g_0)(v_0)|| &\leq ||\rho(g)(v) - \rho(g_0)(v)|| + ||\rho(g_0)(v) - \rho(g_0)(v_0)|| && \text{by the triangular in} \\ &\leq ||\rho(g) - \rho(g_0)||||v|| + ||\rho(g_0)||||v - v_0|| \\ &< \tfrac{\epsilon}{2(||v_0||+\delta)}(||v_0|| + \delta) + ||\rho(g_0)||\delta && \text{since } \begin{cases} ||v|| \leq | \\ \text{the choic} \end{cases} \\ &\leq \epsilon/2 + \epsilon/2 = \epsilon \end{aligned}$$

- given V finite-dimensional, $2 \to 3$

  Let $(e_1, \ldots, e_n)$ be a basis of V and let $||\cdot||$ be the supremum norm. We use the corresponding operator norm on $\mathrm{End}(V)$ and denote it by $||\cdot||$. Let $g_0 \in G$ and $\epsilon > 0$. For every i, the function $\begin{array}{ccc} G & \to & V \\ g & \mapsto & \rho(g)(e_i) \end{array}$ is continuous by 2, so there exists a nhood U_i of g_0 in G s.t. $g \in U_i$ implies $||\rho(g)(e_i) - \rho(g_0)(e_i)|| \leq \epsilon/n$. Let $U = \cap_{i=1}^n U_i$. Then if $g \in U$, then for all $v = \sum_{i=1}^n x_i e_i \in V$, $||\rho(g)(v) - \rho(g_0)(v)|| \leq \sum |x_i| ||\rho(g)(e_i) - \rho(g_0)(e_i)|| \leq \sum |x_i| \epsilon/n \leq \epsilon ||v||$ so $||\rho(g) - \rho(g_0)|| \leq \epsilon$.

A complex Hilbert space is a $\mathbb{C}$-vector space V with a Hermitian inner product s.t. V is complete for the corresponding norm. If V is a finite dimensional $\mathbb{C}$-vector space with a Hermitian inner product, it is complete and therefore a Hilbert space. Write $U(V)$ for the group of unitary endomorphisms of V.

Proposition A.3.2. *If V is a Hilbert space and $\rho : G \to U(V)$ is a morphism of groups, then TFAE*

1. *the map* $\begin{array}{ccc} G \times V & \to & V \\ (g, v) & \mapsto & \rho(g)(v) \end{array}$ *is continuous*

2. $\forall v \in V$, *the map* $\begin{array}{ccc} G & \to & V \\ g & \mapsto & \rho(g)(v) \end{array}$ *is continuous.*

pf.

- by the previous proposition, $2 \to 1$.

- Let $\begin{cases} g_0 \in G \\ v_0 \in V \end{cases}$ and $\epsilon > 0$. Choose a nhood U of g_0 in G s.t. $g \in U$ implies $||\rho(g)(v_0) - \rho(g_0)(v_0)|| < \epsilon/2$ and take $\delta = \epsilon/2$. Then if $g \in U$

$$||v-v_0||<\delta,\quad \begin{aligned}||\rho(g)(v)-\rho(g_0)(v_0)|| &\leq ||\rho(g)(v)-\rho(g)(v_0)||+||\rho(g)(v_0)-\rho(g_0)(v_0)|| \\ &< ||\rho(g)||||v-v_0||+\epsilon/2 \\ &< \epsilon/2+\epsilon/2=\epsilon\end{aligned},$$

since $\rho(g)\in U(V)$.

CLAIM : if $\rho : G \to U(V)$ is a unitary representation of G on a Hilbert space $(V, <\cdot,\cdot>)$, for every G-invariant subspace W of V, the subspace $W^\perp$ is also G-invariant.
pf. if $w \in W^\perp,\ g \in G$, then $\forall v \in W,\ <v, \rho(g)w> = <\rho(g)^{-1}v, w> = 0$ since $\rho(g)^{-1}v \in W$ by the G-invariance of W. This implies $\rho(g)W \in W^\perp$ and $W^\perp$ is G-invariant.
$\to$ if W is a closed G-invariant subspace of V, $V = W \oplus W^\perp$ with $W^\perp$ a closed G-invariant subspace. If V is finite dimensional, every subspace is closed and therefore every unitary representation of G on V is semisimple.

Theorem A.3.2. *Assume that the group G is compact Hausdorff. Let V be a finite-dimensional $\mathbb{C}$-vector space and $\rho : G \to GL(V)$ be a continuous representation of G on V. Then there exists a Hermitian inner product on V that makes ρ a unitary representation.*

pf. Let $<\cdot,\cdot>_0$ be a Hermitian inner product on V. Define $\begin{array}{rcl} <\cdot,\cdot>: V\times V & \to & \mathbb{C} \\ (v,w) & \mapsto & \int_G <\rho(g)v,\rho(g)w>_0\, dg \end{array}$. Then $<\rho(g)v, \rho(g)w> = <v,w>$, $\forall v, w \in V$ and $\forall g \in G$. It is a Hermitian product trivially. Let $v \in V \setminus \{0\}$. Then the function $\begin{array}{rl} G & \to \mathbb{R} \\ g & \mapsto\ <\rho(g)v,\rho(g)v>_0 \end{array}$ is continuous and takes positive values. As G is compact, $\exists \epsilon > 0$ s.t. $<\rho(g)v, \rho(g)v>_0 > \epsilon$, $\forall g \in G$. Then $<v,v> \geq \epsilon > 0$ and it is positive definite.

Corollary A.3.1. *If G is compact Hausdorff, then every finite-dimensional continuous representation of G is semisimple.*

Definition A.3.6. *Let V, W be Hermitian inner product spaces. If $v \in V,\ w \in W$, we define the $\mathbb{C}$-linear map* $\begin{array}{rcl} T^0_{v,w}: V & \to & W \\ x & \mapsto & <x,v>w \end{array}$.

$\to \begin{array}{rcl} V & \to & \hom_{\mathbb{C}}(V,W) \\ v & \mapsto & T^0_{v,w} \end{array}$ is semi-linear.

Proposition A.3.3.

1. *$\forall v \in V,\ \forall w \in W,\ (T^0_{v,w})^* = T^0_{w,v}$*
2. *$\forall v \in V,\ \forall w \in W,\ T^0_{v,w}$ generate $\hom_{\mathbb{C}}(V,W)$ as a $\mathbb{C}$-vector space*
3. *if $V = W,\ \forall v, w \in V,\ T(T^0_{v,w}) = <w,v>$*

4. $\forall v_1, v_2 \in V,\ \forall w_1, w_2 \in W,\ T((T^0_{v_1,w_1} T^0_{v_2,w_2})^*) =< w_1, w_2 >< v_2, v_1 >$

pf.

- $\forall x \in V,\ y \in W,\ < T^0_{v,w}(x), y >=<< x, v > w, y >=< x, v >< w, y >=< x, v > \overline{< y, w >}$
  $=< x, < y, w > v >=< x, T^0_{w,v}(y) >$
- if we choose the orthonormal bases (e_i) of V and (f_j) of W, $\forall i, j$, the matrix of $T^0_{e_i,f_j}$ in these bases is the $m \times n$ with (i, j)-entry equal to 1 and all other entries equal to 0. These clearly generate the $\mathbb{C}$-vector space $M_{mn}(\mathbb{C})$.
- let $(v_1, \ldots, v_n)$ be an orthogonal basis of V s.t. $v_1 = v$. Then $T^0_{v,w}(v_i) = \begin{cases} < v_1, v_1 > w & i = 1 \\ 0 & otherwise \end{cases}$. As $w = \sum \frac{<w,v_i>}{<v_i,v_i>} v_i$, the proposition follows
- $\forall y \in W,\ T^0_{v_1,w_1}(T^0_{v_2,w_2})^* = T^0_{v_1,w_1} T^0_{w_2,v_2}(y) = T^0_{v_1,w_1}(< y, w_2 > v_2) =< y, w_2 >< v_2, v_1 > w_1$. Choose an orthogonal basis $(y_1, \ldots, y_n)$ of W s.t. $y_1 = w_2$. Then $T^0_{v_1,w_1}(T^0_{v_2,w_2}(y))^*(y_i) = \begin{cases} < y_1, y_1 >< v_2, v_1 > w_1 & i = 1 \\ 0 & otherwise \end{cases}$. As $w_1 = \sum \frac{<w_1,y_i>}{<y_i,y_i>} y_i$, the proposition follows.

Let G be a Hausdorff compact group.

Definition A.3.7. *Let $\rho : G \to GL(V)$ be a continuous representation of G on a finite dimensional complex vector space. Remember that the map* $\begin{array}{rcl} \chi_V : G & \to & \mathbb{C} \\ g & \mapsto & T(\rho(g)) \end{array}$ *is called the character of the representation (V, ρ).*

$\to \chi_V : G \to \mathbb{C}$ is a continuous map by the previous proposition A.3.1.

Theorem A.3.3. *Let V, W be continuous representations of G on finite-dimensional complex vector spaces. Assume V, W are both irreducible. Choose G-invariant Hermitian inner products on V, W that will both be denoted by $< \cdot, \cdot >$. Then* $\begin{cases} \forall v_1, v_2 \in V \\ \forall w_1, w_2 \in W \end{cases}$,

$$\int_G < gv_1, v_2 > \overline{< gw_1, w_2 >} dg = \begin{cases} \frac{1}{\dim_{\mathbb{C}} V} < v_1, w_1 > \overline{< v_2, w_2 >} & V \cong W \\ 0 & otherwise \end{cases}$$

pf. If $v \in V, w \in W$, let $T_{v,w} = \int_G g T^0_{v,w} g^{-1} dg \in \hom_{\mathbb{C}}(V, W)$ where $T^0_{v,w}$ is defined above. Then $T_{v,w}$ is G-equivariant, so by the Schur's lemma, $T_{v,w} = \begin{cases} 0 & V \ncong W \\ c(v, w) id_V & V \cong W \end{cases}$ with $c(v, w) \in \mathbb{C}$. Suppose $V \cong W$, then $c(v, w) \dim_{\mathbb{C}} V =$

$T(T_{v,w}) = \int_G T(gT^0_{v,w}g^{-1})dg = T(T^0_{v,w})$ so by the previous proposition, $c(v,w) = \frac{1}{\dim_{\mathbb{C}} V} < w, v >$. Let $I = \int_G < gv_1, v_2 > \overline{< gw_1, w_2 >} dg$. Then $I = \int_G < gv_1, v_2 >< g^{-1}w_2, w_1 > dg = \int_G << g^{-1}w_2, w_1 > gv_1, v_2 > dg = \int_G < gT^0_{w_1,v_1}g^{-1}w_2, v_2 > dg =< T_{w_1,v_1}(w_2), v_2 >$ so $I = \left\{ \begin{array}{rl} 0 & V \ncong W \\ c(w_1,v_1) < w_2, v_2 > & V \cong W \end{array} \right.$ and the theorem follows.

Corollary A.3.2 (Schur's Orthogonality)**.** *Let V, W be continuous representations of G on finite dimensional complex vector spaces. Assume V, W are both irreducible. Then*

$$\int_G \chi_V(g)\overline{\chi_W(g)}dg = \left\{ \begin{array}{ll} 0 & V \ncong W \\ 1 & V \cong W \end{array} \right.$$

pf. Choose G-invariant Hermitian inner products on V, W and fix orthonormal bases $\left\{ \begin{array}{ll} (v_1, \ldots, v_n & \text{for } V \\ (w_1, \ldots, w_n) & \text{for } W \end{array} \right.$. Then for any $\mathbb{C}$-linear endomorphism u of V or W, we have $T(u) = \sum < u(v_i), v_i >$ or $\sum < u(w_j), w_j >$. In particular, $\forall g \in G$, $\left\{ \begin{array}{l} \chi_V(g) = \sum < gv_i, v_i > \\ \chi_W(g) = \sum < gw_j, w_j > \end{array} \right.$ and therefore $\int_G \chi_V(g)\overline{\chi_W(g)}dg = \sum_{i,j} \int_G < gv_i, v_i > \overline{< gw_j, w_j >} dg = \left\{ \begin{array}{rl} 0 & V \ncong W \\ \frac{1}{\dim_{\mathbb{C}} V} \sum_{i,j} < v_i, v_j > \overline{< v_i, v_j >} = 1 & V \cong W \end{array} \right.$ and the corollary follows.

Definition A.3.8. *$L^2(G)$ is the quotient*

$$\{f : G \to \mathbb{C},\ \textit{measurable} \mid \int_G |f(g)|^2 dg < \infty\} / \{f : G \to \mathbb{C},\ \textit{measurable} \mid \int_G |f(g)|^2 dg = 0\}$$

This is a Hilbert space for the Hermitian inner product given by $< f_1, f_2 >= \int_{g \in G} f_1(g)\overline{f_2(g)}dg$.

Definition A.3.9. *If $x \in G$, and f is a function from G to $\mathbb{C}$, define* $\begin{array}{rcl} L_x f : G & \to & \mathbb{C} \\ g & \mapsto & f(xg) \end{array}$, $\begin{array}{rcl} R_x f : G & \to & \mathbb{C} \\ g & \mapsto & f(gx) \end{array}$.

Proposition A.3.4. *$\forall x, y \in G$, the operators L_x, R_x induce endomorphisms L_x, R_x of $L^2(G)$ with the following properties;* $\begin{array}{r} L_x R_y = R_y L_x,\ \ R_x R_y = R_{xy},\ \ L_x L_y = L_{yx} \\ R_x^* = R_x^{-1} = R_{x^{-1}},\ \ L_x^* = L_x^{-1} = L_{x^{-1}} \end{array}$. *In particular, R_x, L_x are unitary endomorphisms of $L^2(G)$.*

pf.

- Let $f_1, f_2 : G \to \mathbb{C}$ be measurable functions. Then $\left\{ \begin{array}{ll} < R_x f_1, R_x f_2 >= \int_G f_1(gx)\overline{f_2(gx)}dg = \int_G f_1(g)\overline{f_2(g)}dg =< f_1, f_2 > & \text{by the right invarianc} \\ < L_x f_1, L_x f_2 >= \int_G f_1(xg)\overline{f_2(xg)}dg = \int_G f_1(g)\overline{f_2(g)}dg =< f_1, f_2 > & \text{by the left-invarian} \end{array} \right.$ Thus R_x, L_x preserves both spaces in the quotient defining $L^2(G)$ and therefore they induce endomorphisms of $L^2(G)$.

- Let $f \in L^2(G),\ \ g \in G$. Then $\begin{cases} (L_xR_yf)(g) = (R_yf)(xg) = f(xgy) = (L_xf)(gy) = (R_yL_xf)(g \\ (L_xL_yf)(g) = (L_yf)(xg) = f(yxg) = (L_{yx}f)(g \\ (R_xR_yf)(g) = (R_yf)(gx) = f(gyx) = (R_{xy}f)(g \end{cases}$

- by the first stmt, L_x, R_x are unitary endomorphisms and by the second stmt, $L_xL_{x^{-1}} = L_{xx^{-1}} = L_1 = Id \to L_x^{-1} = L_{x^{-1}}$ and similarly for R_x.

Definition A.3.10. *If f is a function $G \to \mathbb{C}$, define a function* $\begin{array}{rcl} \tilde{f} : G & \to & \mathbb{C} \\ g & \mapsto & \overline{f(g^{-1})} \end{array}$.

Proposition A.3.5. *The above function defines a unitary endomorphism $f \mapsto \tilde{f}$ of $L^2(G)$.*

pf. Let $f : G \to \mathbb{C}$ be a measurable function. Then, $\int_G \overline{|f(g^{-1})|}^2 dg = \int_G |f(g)|^2 dg$ because G being compact implies the Haar measure dg is equal to its own pullback by $g \mapsto g^{-1}$. Then the proposition follows.

Definition A.3.11. *If $f_1, f_2 \in L^2(G)$, define* $\begin{array}{rcl} f_1 * f_2 : G & \to & \mathbb{C} \\ g & \mapsto & \int_G f_1(gx)f_2(x^{-1})dx \end{array}$, *called the convolution product of f_1, f_2.*

Proposition A.3.6. *1. $\forall f_1, f_2 \in L^2(G),\ \ f_1 * f_2$ is a bounded function on G and we have $||f_1 * f_2||_\infty \leq ||f_1||_2||f_2||_2$*

2. the convolution product is associative and distributive w.r.t. addition

pf.

- $\forall g \in G,\ \ |f_1 * f_2(g)| = |\int_G f_1(gx)f_2(x^{-1})dx| = |\int_G (L_gf_1)\overline{\tilde{f_2}(x)}dx| \leq ||L_gf_1||_2||\tilde{f_2}||_2 = ||f_1||_2||f_2||_2$ by CS inequality.

- $\forall f_1 f_2, f_3 \in L^2(G),\ \ \forall g \in G$
$\begin{cases} ((f_1 * f_2) * f_3)(g) = \int_G (f_1 * f_2)(gx)f_3(x^{-1})dx = \int_G(\int_G f_1(gxy)f_2(y^{-1})dy)f_3(x^{-1})dx \\ (f_1 * (f_2 * f_3))(g) = \int_G f_1(gz)(f_2 * f_3)(z^{-1})dz = \int_G f_1(gz)(\int_G f_2(z^{-1}t)f_3(t^{-1})dt)dz \end{cases}$.
By the change of variables $\begin{cases} z \mapsto y \\ z \mapsto y^{-1}x \end{cases}$ and Fubini's theorem and bi-invariance Haar measure dg, they are equivalent, hence it is associative.

$\to f_1 * f_2$ are continuous for all $f_1, f_2 \in L^2(G)$ since if f_1, f_2 are the limit in $L^2(G)$ of sequences $f_{(1,n)}, f_{(2,n)}$, then the sequence $(f_{1,n} * f_{2,n}) \to f_1 * f_2$ in $L^\infty(G)$. Since the continuous functions are dense in $L^2(G)$ and the convolution of the two continuous functions is continuous, the continuity follows.

Definition A.3.12. *Let $f \in L^2(G)$. Define endomorphisms L_f, R_f of $L^2(G)$ by* $\begin{cases} L_f(f_1) = f * f_1 \\ R_f(f_1) = f_1 * f \end{cases}$.

$\rightarrow$ well-defined and continuous by the previous proposition.

Proposition A.3.7. *1.* $\forall f \in L^2(G),\ R_f^* = R_{\tilde{f}},\ L_f^* = L_{\tilde{f}}$.

2. $\forall f \in L^2(G),\ \ \forall x \in G,\ R_x L_f = L_f R_x,\ \ R_f L_x = L_x R_f$

pf.

- $\forall f_1, f_2 \in L^2(G)$,
$$\begin{array}{rlr} < R_f(f_1), f_2 >= \int_G (f_1 * f)\overline{f_2(g)}dg & = \int_G \int_G f_1(gx^{-1})f(x)\overline{f_2(g)}dxdg & \\ = \int\int f_1(y)\overline{f_2(yx)}f(x)dxdy & \text{by the change of variables } y = gx^{-1} & \\ = \int\int f_1(y)\overline{f_2(yz^{-1})}\overline{\tilde{f}(z)}dzdy & \text{by the change of variables } z = x^{-1} & . \\ = \int_G f_1(y)\overline{(f_2 * \tilde{f})(y)}dy =< f_1, R_{\tilde{f}} f_2 > & & \end{array}$$
  Similarly for L_f^*.

- $\forall h \in L^2(G),\ \ \forall g \in G,\ \left\{ \begin{array}{r} (R_x L_f(h))(g) = (f * h)(gx) = \int_G f(gxy)h(y^{-1})dy \\ (L_f R_x(h))(g) = (f * (R_x h))(g) = \int_G f(gz)h(z^{-1}x)dz \end{array} \right.$
  are equivalent by the change of variables $z = xy$.
  Similarly for $R_f L_x = L_x R_f$.

Definition A.3.13. *If V_1, V_2 are two Hilbert spaces over $\mathbb{C}$ and $T : V_1 \rightarrow V_2$ is a continuous $\mathbb{C}$-linear map, we say T is compact if the set $\overline{T(B)} \subset V_2$ is compact, where $B = \{v \in V_1 | ||v|| = 1\}$.*

Lemma A.3.1. *1. every finite rank operator is compact*

2. the space of compact operators $T : V_1 \rightarrow V_2$ is closed in $\hom(V_1, V_2)$

pf.

- The closed unit ball of a finite dimensional $\mathbb{C}$-vector space is compact, and therefore every finite rank operator is compact.

- Let (T_n) be a sequence of compact operators in $\hom(V_1, V_2)$ and suppose it converges to $T \in \hom(V_1, V_2)$. Let (x_n) be a sequence in B. Construct a sequence $x^{(r)} = (x_n^{(r)})$ of subsequences of (x_n) in the following inductive way; take $x^{(0)} = (x_n)$. If $r \geq 1$, suppose x^{r-1} is constructed and take for $x^{(r)}$ a subsequence of $x^{(r-1)}$ s.t. $(T_r(x_n^{(r)})$ is a Cauchy sequence. Set $y_n = x_n^{(n)}$. Let $\epsilon > 0$. Choose $m \geq 1$ s.t. $||T - T_m|| \leq \epsilon$ and $N \geq m$ s.t. $\forall r, s \geq N,\ \ ||T_m(y_r) - T_m(y_s)|| \leq \epsilon$. Then if $r, s \geq N,\ \ ||T(y_r) - T(y_s)|| \leq ||(T - T_m)(y_r)|| + ||T_m(y_r - y_s)|| + ||(T_m - T)(y_s)|| \leq 3\epsilon$, given $y_s, y_r \in B$. Hence $(T(y_n))$ is a Cauchy sequence. Since V_2 is complete, the lemma follows.

Theorem A.3.4. *Let V_1, V_2 be Hilbert spaces. Write* $\hom(V_1, V_2)$ *for the space of continuous $\mathbb{C}$-linear maps from V_1 to V_2 and consider the topology on it defined by the operator norm. Then $\forall T \in \hom(V_1, V_2)$, TFAE*

1. *T is compact*
2. *T is a limit of finite rank element of* $\hom(V_1, V_2)$

pf.

- $2 \to 1$ follows by the previous lemma.
- $1 \to 2$

  pf. Let $T \in \hom(V_1, V_2)$ be a compact operator. Let (W_n) be a sequence of finite dimensional subspace of V_2 s.t. $V_2 = \cup W_n$, and $\forall n \geq 0,\ \pi_n : V_2 \to W_n$ be the orthogonal projection. Set $T_n = \pi_n \circ T$. The $\forall x \in V_1,\ T_n(x) \to T(x)$ as $n \to \infty$. Let $\epsilon > 0$. As $\overline{T(B)}$ is compact, $\exists x_1, \ldots, x_r \in B$ s.t. $\forall y \in B,\ \exists i \in \{1, \ldots, r\}$ s.t. $||T(x_i) - T(y)|| \leq \epsilon$. Then $\forall n \geq 0,\ ||T_n(y) - T_n(x_i)|| = ||\pi_n(T(y) - T(x_i))|| \leq ||T(y) - T(x_i)|| \leq \epsilon$. Choose $N \geq 0$ s.t. $\forall i,\ \forall n \geq N,\ ||T_n(x_i) - T(x_i)|| \leq \epsilon$. Then fix $n \geq N$, and if $y \in B$ choose $i \in \{1, \ldots, r\}$ s.t. $||T(y) - T(x_i)|| \leq \epsilon$. Then $||T(y) - T_n(y)|| \leq ||T(y) - T(x_i)|| + ||T(x_i) - T_n(x_i)|| + ||T_n(x_i) - T_n(y)|| \leq 3\epsilon$.

Definition A.3.14. *Let V be a Hilbert space, and T be a continuous endomorphism of V. We say $\lambda \in \mathbb{C}$ is an eigenvalue of T if $\ker(T - \lambda id_V) \neq \{0\}$. We denote by $Spec(T) \subset \mathbb{C}$ the set of eigenvalues of T and call it the spectrum of T.*

Theorem A.3.5 (Spectral Theorem)**.** *Let V be a Hilbert space over $\mathbb{C}$, and let $T : V \to V$ be a continuous endomorphism of V. Assume T is compact and self-adjoint, and write $V_\lambda = \ker(T - \lambda id_V),\quad \forall \lambda \in \mathbb{C}$. Then*

$$\begin{cases} Spec(T) \subset \mathbb{R} \\ \text{if } \lambda, \mu \in Spec(T),\ \lambda \neq \mu, \text{ then } V_\mu \subset V_\lambda^\perp \\ \text{if } \lambda \in Spec(T) \setminus \{0\},\ \dim_{\mathbb{C}} V_\lambda < \infty \\ Spec(T) \text{ is finite or countable} \\ \bigoplus_{\lambda \in Spec(T)} V_\lambda \text{ is dense in } V \end{cases}$$

Theorem A.3.6. *$\forall f \in L^2(G),\quad \exists (f_n)_{n \geq 0}$ a sequence of functions in $L^2(G)$ s.t.*

$$\begin{cases} \tilde{f}_n = f_n,\ \forall n \geq 0 \\ ||f_n||_2 = 1,\ \forall n \geq 0 \\ f * f_n \to f \text{ in } L^2(G) \text{ as } n \to \infty \end{cases}.$$

pf. Let $\epsilon > 0$. Choose a nhood U of 1 in G s.t. $U = U^{-1}$ and $\forall x, y \in U,\ ||R_x f - f||_2 < \epsilon$. Let $h_\epsilon = \frac{1}{vol(U)} \not\Vdash_U$. Then $\tilde{h}_\epsilon = h_\epsilon$ and $||h_\epsilon||_2 = 1$. Also $(f * h_\epsilon)(g) = \int_G f(gx^{-1}) h_\epsilon(x) dx = \frac{1}{vol(U)} \int_U f(gx^{-1}) dx$

$\to (f*h_\epsilon)(g) - f(g) = \frac{1}{vol(U)}\int_U (R_{x^{-1}}(f)(g) - f(g))dx$. This gives $||(f*h_\epsilon)-f||_2^2 = \int_G |(f*h_\epsilon)(g)-f(g)|^2 dg = \frac{1}{vol(U)^2}\int_{G\times U\times U}(R_{x^{-1}}f-f)(g)\overline{R_{y^{-1}}f - f(g)}dxdydg$
$= \frac{1}{vol(U)^2}\int_{U\times U} < R_{x^{-1}}f - f, R_{y^{-1}}f - f > dxdy \le \frac{1}{vol(U)^2}\int ||R_{x^{-1}}f - f||_2||R_{y^{-1}}f - f||_2 dxdy$
$\le \frac{1}{vol(U)^2}\int \epsilon^2 dxdy = \epsilon^2$ by the CS inequality and the choice of U. If we take $f_n = h_{1/2^n}$, the theorem follows.

Lemma A.3.2. *If $f_1, f_2 : G \to \mathbb{C}$, define* $\begin{array}{rcl} f_1\otimes f_2 : G\times G & \to & \mathbb{C} \\ (g_1,g_2) & \mapsto & f_1(g_1)f_2(g_2) \end{array}$. *This induces a $\mathbb{C}$-linear map $L^2(G)\otimes_{\mathbb{C}} L^2(G) \to L^2(G\times G)$ which is injective with dense image.*

pf. Take the Hilbert basis $(e_i)_{i\in I}$ of $L^2(G)$. Then the family $(e_i\otimes e_j)$ of $L^2(G\times G)$ is orthonormal and any function on $L^2(G\times G)$ that is orthogonal to all $e_i\otimes e_j$ is 0 a.e. hence 0.

Lemma A.3.3. *$\forall K \in L^2(G\times G)$ define* $\begin{array}{rcl} T_K : L^2(G) & \to & L^2(G) \\ f(x) & \mapsto & \int_G K(x,y)f(y)dy \end{array}$. *Then $||T_K|| \le ||K||_2$.*

Theorem A.3.7. *$\forall f \in L^2(G)$, the endomorphisms R_f and L_f of $L^2(G)$ are compact.*

pf. Consider $\begin{array}{rcl} K : G\times G & \to & \mathbb{C} \\ (x,y) & \mapsto & f(xy^{-1}) \end{array}$. Then $K \in L^2(G\times G)$, so by the previous lemma, $\exists (g_i)_{i\in I},\ (h_i)_{i\in I}$ in $L^2(G)$ s.t. $\sum g_i\otimes h_i$ converges to K in $L^2(G\times G)$. Here $T_K = L_f$. For every finite subset J of I, let $S_J = \sum_{i,j\in J} g_i\otimes h_j \in L^2(G\times G)$ and $T_J = T_{S_J}$. Then $\forall h \in L^2(G)$, $\forall x \in G$, $(T_Jh)(x) = \sum_{(i,j)\in J^2}\int_G h(y)g_i(x)h_j(y)dy = \sum_i\sum_j\int_G h(y)h_j(y)dy g_i(x)$, i.e. $\forall h \in L^2(G)$, T_Jh is in the finite-dimensional subspace of $L^2(G)$ spanned by $g_i,\ i\in J$. Hence the operator T_J has finite rank. By the previous lemma, $||T_J||_2 \le ||S_J||_2 \to ||K||_2$ so L_f is the limit of the operator T_J and therefore L_f is compact by the previous theorem.

Definition A.3.15. *We make $G\times G$ act on $L^2(G)$ by $(x,y)\cdot f = L_{x^{-1}}R_yf = R_yL_{x^{-1}}f$, i.e. $((x,y)\cdot f)(g) = f(x^{-1}gy),\ \forall g\in G$.*

Proposition A.3.8. *The representation of $G\times G$ on $L^2(G)$ defined above is continuous and so it is a unitary representation.*

pf. $L_{x^{-1}}R_y$ is a unitary endomorphism of $L^2(G)$, $\forall x,y\in G$, so if the representation is continuous, the proposition follows. Fix $f_0 \in L^2(G)$ and let $x_0, y_0 \in G$, and $\epsilon > 0$. Choose $f : G \to \mathbb{C}$ continuous s.t. $||f_0 - f||_2 \leq \epsilon/3$. As G is compact and f is continuous, f is uniformly continuous, so there exists a nhood U of 1 in G s.t. $\forall x, y \in U,\ \forall g \in G,\ |f(x^{-1}gy) - f(g)| \leq \epsilon/3$. If $x \in x_0U,\ y \in y_0U,\ ||L_{x^{-1}}R_y f - L_{x_0^{-1}}R_{y_0} f||_2 = \sqrt{\int_G |f(x^{-1}gy) - f(x_0^{-1}gy_0)|^2 dg} \leq \epsilon/3$. Then $||L_{x^{-1}}R_y f_0 - L_{x_0^{-1}}R_{y_0} f_0||_2 \leq ||L_{x^{-1}}R_y f_0 - L_{x^{-1}}R_y f||_2 + |L_{x^{-1}}R_y f - L_{x_0^{-1}}R_{y_0} f||_2 + ||L_{x_0^{-1}}R_{y_0} f - L_{x_0^{-1}}R_{y_0} f_0||_2 \leq \epsilon$, so the map $\begin{array}{ccc} G \times G & \to & L^2(G) \\ (x, y) & \mapsto & L_{x^{-1}}R_y f_0 \end{array}$ is continuous. Then by the previous proposition A.3.1., the representation is continuous.

Assume G is a compact Hausdorff group.

Definition A.3.16. *Denote by $\hat{G}$ the set of isomorphism classes of irreducible continuous representations of G on finite dimensional complex vector spaces.*

Let $(V_\rho, \rho) \in \hat{G}$. Fix a G-invariant Hermitian inner product $< \cdot, \cdot >$ on V_ρ. Then $G \times G$ acts on $\mathrm{End}_{\mathbb{C}}(V_\rho) \cong V_\rho^* \otimes_{\mathbb{C}} V_\rho$ by $(g_1, g_2) \cdot u = \rho(g_1)u\rho(g_2)^{-1}$.

Definition A.3.17. *The Hilbert-Schmidt inner product on $End_{\mathbb{C}}(V_\rho)$ is given by the formula*

$$< u, v >_{HS} = T(uv^*)$$

where v^* is the adjoint of v for the chosen G-invariant inner product $< \cdot, \cdot >$ on V_ρ.

$\to$Hermitian inner product and $G \times G$-invariant given $\rho(g)$ begin unitary $\forall g \in G$.

Lemma A.3.4. *Let V be a Hilbert space with a unitary representation of G and let $V_1, V_2 \subset V$ be finite-dimensional irreducible subrepresentations of G. Then either $V_1 \cong V_2$ as representations of G or $< v_1, v_2 >= 0,\ \forall v_1 \in V_1,\ v_2 \in V_2$.*

pf. $V_2^\perp$ is stable by G and $V = V_2 \oplus V_2^\perp$. So the orthogonal projection $\pi : V \to V_2$ is G-equivariant, hence the composition $V_1 \subset V \to V_2$ is G-equivariant. If $V_1 \ncong V_2$, then this G-equivariant map $V_1 \to V_2$ has to be 0 by Schur's lemma, which means $V_1 \subset V_2^\perp$.

Define $\begin{array}{rcl} \iota_\rho : \mathrm{End}_{\mathbb{C}}(V_\rho) & \to & L^2(G) \\ u & \mapsto & (g \mapsto T(\rho(g)^{-1} \circ u)) \end{array}$ ($\forall u \in \mathrm{End}_{\mathbb{C}}(V_\rho)$, the function $g \mapsto T(\rho(g)^{-1} \circ u)$ is continuous on G, hence on $L^2(G)$, given G being compact. So it is well-defined).

Theorem A.3.8. 1. *$\forall (V_\rho, \rho) \in \hat{G}$, $End_{\mathbb{C}}(V_\rho)$ is an irreducible representation of $G \times G$. Also if $(V_\rho, \rho) \ncong (V_{\rho'}, \rho')$, then $End_{\mathbb{C}}(V_\rho) \ncong End_{\mathbb{C}}(V_{\rho'})$*

2. *$\forall (V_\rho, \rho) \in \hat{G}$, the map $\iota_\rho : End_{\mathbb{C}}(V_\rho) \to L^2(G)$ is $G \times G$-equivariant, the map $\dim(V_\rho)^{1/2}\iota_\rho$ is an isometry, and $(\dim_{\mathbb{C}} V_\rho)\iota_\rho$ sends the composition in $End_{\mathbb{C}}(V_\rho)$ to the convolution product in $L^2(G)$*

3. *the map $\iota = \bigoplus_{\rho\in\hat{G}} \iota_\rho : \bigoplus_{\rho\in\hat{G}} End_{\mathbb{C}}(V_\rho) \to L^2(G)$ is injective and has dense image, i.e. as a representation of $G \times G$ and as $\mathbb{C}$-algebra,*

$$L^2(G) \cong \widehat{\bigoplus_{\rho\in\hat{G}} End_{\mathbb{C}}(V_\rho)}$$

pf.

- let $\rho \in \hat{G}$. Then $\chi_{\mathrm{End}_{\mathbb{C}}(V_\rho)}(g_1, g_2) = \chi_{V_\rho}(g_1)\overline{\chi_{V_\rho}(g_2)}$ by the definition of the action of $G \times G$ on $\mathrm{End}_{\mathbb{C}}(V_\rho)$. By the Schur's orthogonality formula, $\int_{G\times G} |\chi_{\mathrm{End}_{\mathbb{C}}(V_\rho)}(g_1, g_2)|^2 dg_1 dg_2 = (\int_G |\chi_{V_\rho}(g_1)|^2 dg_1)(\int_G |\chi_{V_\rho}(g_2)|^2 dg_2) = 1$ so $\mathrm{End}_{\mathbb{C}}(V_\rho)$ is irreducible. Moreover, if $(V_{\rho'}, \rho') \ncong (V_\rho, \rho)$, then by the Schur's orthogonality, $< \chi_{\mathrm{End}_{\mathbb{C}}(V_\rho)}, \chi_{\mathrm{End}_{\mathbb{C}}(V_{\rho'})} >_{L^2(G\times G)} =< \chi_{V_\rho}, \chi_{V_{\rho'}} >_{L^2(G)} < \overline{\chi_{V_\rho}}, \overline{\chi_{V_{\rho'}}} >_{L^2(G)} = 0$ so $\mathrm{End}_{\mathbb{C}}(V_\rho) \ncong \mathrm{End}_{\mathbb{C}}(V_{\rho'})$.

- fix $(V_\rho, \rho) \in \hat{G}$.
  $\forall x, y \in G, \ \ \forall u \in \mathrm{End}_{\mathbb{C}}(V_\rho), \ \ \forall g \in G,$
  $\iota_\rho(\rho(x)u\rho(y)^{-1})(g) = T(\rho(g)^{-1}\rho(x)u\rho(y)^{-1}) = T(\rho(x^{-1}gy)^{-1}u) = (L_{x^{-1}}R_y\iota_\rho(u))(g) \to$ ι_ρ is $G \times G$-equivariant.
  Let $u_1, u_2 \in \mathrm{End}_{\mathbb{C}}(V_\rho)$. $\mathrm{End}_{\mathbb{C}}(V_\rho)$ is spanned by the $T^0_{v,w}$, $\forall v, w \in V_\rho$, so assume $\begin{cases} u_1 = T^0_{v_1,w_1} \\ u_2 = T^0_{v_2,w_2} \end{cases}$, $\exists v_1, v_2, w_1, w_2 \in V_\rho$. Then,
  $$\begin{aligned} < \iota_\rho(u_1), \iota_\rho(u_2) > &= \int_G T(\rho(g)^{-1}T^0_{v_1,w_1})\overline{T(\rho(g)^{-1}T^0_{v_2,w_2})}dg & \rho(g)T^0_{v,w} = T^0_{v,gw} \\ &= \int_G < g^{-1}v_1, w_1 > \overline{< g^{-1}w_2, v_2 >}dg \\ &= \int_G < gv_2, w_2 > \overline{< gv_1, w_1 >}dg \\ &= \tfrac{1}{\dim_{\mathbb{C}} V_\rho} < v_2, v_1 >< w_1, w_2 > & \text{by the theorem A.3.3.} \\ &= \tfrac{1}{\dim_{\mathbb{C}} V_\rho} < u_1, u_2 >_{HS} & \text{by the proposition A.3.3.} \end{aligned}$$
  Thus $\dim(V_\rho)^{1/2}\iota_\rho$ is an isometry.
  Let $g \in G$. Then $\iota_\rho(u_1u_2)(g) = T(\rho(g)^{-1}T^0_{v_1,w_1}(T^0_{w_2,v_2})^*) =< \rho(g)^{-1}w_1, v_2 >< w_2, v_1 >$ by the proposition A.3.3. On the other hand,
  $$\begin{aligned} (\iota_\rho(u_1) * \iota_\rho(u_2))(g) &= \int_G T(\rho(x)^{-1}\rho(g)^{-1}T^0_{v_1,w_1})T(\rho(x)T^0_{v_2,w_2})dx \\ &= \int_G < \rho(x)^{-1}\rho(g)^{-1}w_2, v_1 >< \rho(x)w_2, v_2 > dx \\ &= \int_G < \rho(x)w_2, v_2 > \overline{< \rho(x)v_1, \rho(g)^{-1}w_1 >}dx \\ &= \tfrac{1}{\dim_{\mathbb{C}}(V_\rho)} < \rho(g)^{-1}w_1, v_2 >< w_2, v_1 > & \text{by the theorem A.3.3.} \end{aligned}$$

- Let $u = (u_\rho) \in \bigoplus_{\rho\in\hat{G}} \mathrm{End}_{\mathbb{C}}(V_\rho)$ be s.t. $\iota(u) = 0$. Then $\sum \iota_\rho(u_\rho) = 0$. By the previous lemma, and the statement 1, the subspaces $\iota_\rho(\mathrm{End}_{\mathbb{C}}(V_\rho)$

are pairwise orthogonal to each other, so $\forall \rho \in \hat{G},\ 0 =< \iota_\rho(u_\rho), \iota(u) >=< \iota_\rho(u_\rho), \iota_\rho(u_\rho) >= \frac{1}{\dim_{\mathbb{C}} V_\rho} < u_\rho, u_\rho >$. Thus all $u_\rho = 0$, i.e. $u = 0$ and ι is injective.

CLAIM : Every finite dimensional G-subrepresentation of $L^2(G)$ is contained in Im(ι), where we make G act on $L^2(G)$ by the left regular action.

pf. As finite dimensional representatons of G are semisimple, if $\forall \rho \in \hat{G},\ \forall u : V_\rho \to L^2(G),\ G$-equivariant map, if $\mathrm{Im}(u) \subset \mathrm{Im}(\iota)$, then the claim follows. Fix $\rho \in \hat{G}$ and $u : V_\rho \to L^2(G),\ G$-equivariant map. Let $v \in V_\rho$. Then $\forall f \in L^2(G),\ \forall g \in G,$

$$\begin{aligned}
(u(v) * f)(g) &= \int_G u(v)(gx) f\overline{(x^{-1})} dx \\
&= \int_G L_g(u(v))(x) \tilde{f}(x) dx \\
&=< L_g(u(v)), \tilde{f} >_{L^2(G)} \\
&=< u(\rho(g)^{-1}v), \tilde{f} >_{L^2(G)} && \text{by the } G\text{-equivariance of } u \ . \\
&=< \rho(g)^{-1}v, u^*(\tilde{f}) >_{V_\rho} \\
&= T(\rho(g)^{-1}T_{u^*(\tilde{f}),v}) && \text{by the definition of } T^0_{v,w} \\
&= \iota_\rho(T^0_{u^*(\tilde{f}),v})(g)
\end{aligned}$$

Thus $u(v) * f \in \mathrm{Im}(\iota_\rho),\ \ \forall f \in L^2(G)$. Then there exists a sequence (f_n) in $L^2(G)$ s.t. $u(v) * f_n \to u(v)$ as $n \to \infty$ by the theorem A.3.6. Thus $u(v) \in \overline{\mathrm{Im}\iota_\rho}$. As $\mathrm{Im}\iota_\rho$ is a finite-dimensional subspace of $L^2(G)$, it is closed in $L^2(G)$ so $u(v) \in \mathrm{Im}\iota_\rho \subset \mathrm{Im}\iota$.

Let $f \in \mathrm{Im}(\iota)^\perp$. Then f is orthogonal to every finite dimensional G-subrepresentation of $L^2(G)$. Let $h \in L^2(G)$ be s.t. $h = \tilde{h}$. Then the endomorphism R_h of $L^2(G)$ is self-adjoint and compact by the theorem A.3.7. Hence by the spectral theorem, $\ker R_h$ is the orthogonal of the closure of the direct sum of the eigenspaces $\ker(R_h - \lambda id_{L^2(G)}),\ \lambda \in \mathbb{R}^\times$, which are all finite dimensional. As R_h is G-equivariant and commutes with the left regular action, there eigenspaces are left stable by G and so f is orthogonal to all of them, hence $f \in \ker R_h,\ \ f * h = 0$. Then by the theorem A.3.6., there exists a sequence $(h_n)_{n \geq 0}$ of elements of $L^2(G)$ s.t. $\tilde{h}_n = h_n,\ \ \forall n \geq 0$ and $f * h_n \to f$ as $n \to \infty$. Then $f * h_n = 0,\ \forall n \geq 0$ and therefore $f = 0$. So $\mathrm{Im}\iota$ is dense in $L^2(G)$.